Vector Integeration

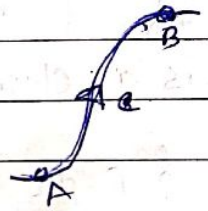
★ Line Integral :

Let $\bar{f}$ be a vector point function defined at every point (x,y,z) on curve C.

If $\bar{r}$ is a position vector of a point $P(x,y,z)$.

i.e. $\bar{r} = x\,i + y\,j + z\,k$

Then $d\bar{r} = dx\,i + dy\,j + dz\,k$



Thus the line integral of vector $\bar{f} = f_1\,\hat{i} + f_2\,\hat{j} + f_3\,\hat{k}$ over a curve C is defined as

$$\int_C \bar{F}\cdot d\bar{r} = \int_C (f_1\,dx + f_2\,dy + f_3\,dz)$$

Note (1)

If the curve C is represented by a parametric eqns

i.e. $x = x(t)$, $y = y(t)$, $z = z(t)$.

Then

$$\bar{r} = x(t)\,\hat{i} + y(t)\,\hat{j} + z(t)\,\hat{k}$$

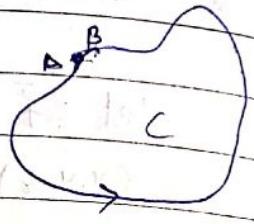
$$\Rightarrow \frac{d\bar{r}}{dt} = \frac{dx}{dt}\,i + \frac{dy}{dt}\,j + \frac{dz}{dt}\,k$$

Therefore a line integral of vector function $\bar{f} = f_1\,i + f_2\,j + f_3\,k$ from $t=a$ to $t=b$ is given by

$$\int_C F\,dr = \int_C F\,\frac{dr}{dt}\,dt = \int_{t=a}^{t=b} \left(f_1\,\frac{dx}{dt} + f_2\,\frac{dy}{dt} + f_3\,\frac{dz}{dt}\right) dt$$

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Note (iii) If C is a closed curve, then the symbol of line integral $\int_C$ is replaced by $\oint_C$



✱ Circulation :

If $\bar{f}$ is a velocity of a fluid particle and C is a closed curve (contour) then the line integral $\oint_C \bar{F} \cdot d\bar{r}$ represented as circulation of $\vec{F}$ around curve C.

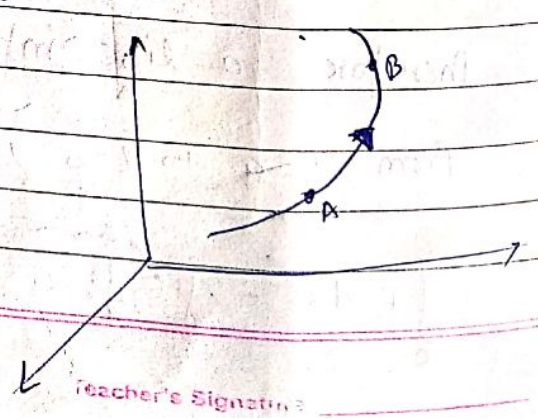
Thus if $\oint_C F \cdot dr = 0$ i.e. circulation is zero

$\Rightarrow \vec{F}$ is Irrotational (Irrotational) vector point function (vector field)

✱ Workdone by a force :

If $\bar{F}$ is the force acting on a particle moving along the arc AB of the curve C, then the line integral $\int_A^B \vec{F} \cdot d\bar{r}$ represent the work done in moving (displacing) particle from A to B.

Note
Direction of C curve should be taken as anticlockwise



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✱ Path independence of line Integral :

If $\vec{F}$ is conservative vector field.

i.e $\vec{F} = \nabla\phi$, where ϕ is scalar potential,

then line integral from A to B along the curve C. is

$$\int_C \bar{F}\cdot d\bar{r} = \int_A^B \nabla\phi \cdot d\bar{r}$$

$$= \int_A^B \left(\frac{\partial\phi}{\partial x}i + \frac{\partial\phi}{\partial y}j + \frac{\partial\phi}{\partial z}k\right)\cdot\left(dx\,i + dy\,j + dz\,k\right)$$

$$= \int_A^B \left(\frac{\partial\phi}{\partial x}\cdot dx + \frac{\partial\phi}{\partial y}\cdot dy + \frac{\partial\phi}{\partial z}dz\right)$$

$$= \int_A^B d\phi \quad \because \text{(By theorem of total differentiation)}$$

$$= \phi(B) - \phi(A)$$

Thus the line integral for conservative vector field is depend on only initial and terminal points ie. it is path independence.

Note (1) If $\bar{F}$ is conservative and C is a closed curve $\because$ ie $(A=B)$

then $$\oint_C \bar{F}\cdot d\bar{r} = \phi(A) - \phi(A) = 0$$

(2) If $\bar{F}$ is conservative, then

$$\nabla \times \bar{F} = \nabla \times (\nabla\phi) = 0$$

Hence conservative field is irrotational.

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Ex: Evaluate $\int_C \bar{F} \cdot d\bar{r}$ where $\bar{F} = (x^2+y^2)\,\hat{i} - 2xy\,\hat{j}$ and C is the rectangle in xy-plane bounded by $y=0,\ x=a,\ y=b,\ x=0$

Soln. Given that

$$\bar{F} = (x^2+y^2)\,\hat{i} - 2xy\,\hat{j} + 0\,\hat{k} \quad \text{---(1)}$$

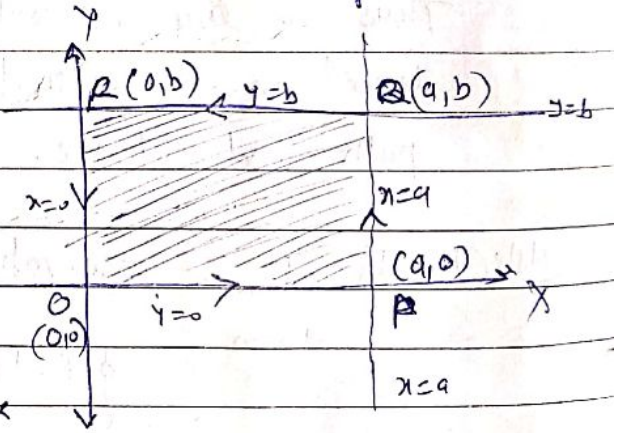
let $\bar{r} = x\,\hat{i} + y\,\hat{j} + z\,\hat{k}$

$\therefore d\bar{r} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$

$$\therefore \bar{F} \cdot d\bar{r} = \left((x^2+y^2)\,\hat{i} - 2xy\,\hat{j} + 0\,\hat{k}\right) \cdot \left(dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}\right)$$

$$= (x^2+y^2)\,dx + (-2xy)\,dy + 0 \quad \text{---(2)}$$

Now let C be the rectangle OPQR bounded by four lines $y=0,\ x=a,\ y=b,\ x=a$ as shown in fig.



Thus

$$\int_C \bar{F} \cdot d\bar{r} = \int_{OPQR} \bar{F} \cdot d\bar{r}$$

$$= \int_{OP} \bar{F} \cdot d\bar{r} + \int_{PB} \bar{F} \cdot d\bar{r} + \int_{QR} \bar{F} \cdot d\bar{r} + \int_{RO} \bar{F} \cdot d\bar{r}$$

---(3)

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Now along line OP

$y=0 \Rightarrow dy=0$

and x varies from $x=0$ to $x=a$

$$\therefore \int_{OP} \bar{F}\cdot d\bar{r} = \int_{x=0}^{a} (x^2+0)\,dx + (-2x\cdot 0)\cdot 0 = \int_0^a x^2\,dx$$

$$= \left[x^3/3\right]_0^a = a^3/3.$$

Along PQ.

$x=a \Rightarrow dx=0$

and y varies from $y=0$ to $y=b$

$$\therefore \int_{PQ} \bar{F}\cdot d\bar{r} = \int_{y=0}^{b} (0 + (-2ay)\,dy) = \int_{y=0}^{b} -2ay\,dy$$

$$= -2a\left[y^2/2\right]_0^b = -ab^2.$$

Along QR

$y=b \Rightarrow dy=0$

and x varies from a to 0

$$\therefore \int_{QR} \bar{F}\cdot d\bar{r} = \int_{x=a}^{0} (x^2+b^2)\,dx + (-2xb)\cdot 0 = \int_a^0 (x^2+b^2)\,dx$$

$$= \left[\frac{x^3}{3} + b^2x\right]_a^0 = \left(0 - \frac{a^3}{3} - ab^2\right)$$

$$= -\left(\frac{a^3}{3} + ab^2\right)$$

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along line RO

ie $x = 0 \Rightarrow dx = 0$

and y varies from $y=b$ to $y=0$

$$\therefore \int_{RO} \bar{F} \cdot d\bar{r} = \int_{y=b}^{0} (0 + 0\, dy) = 0$$

Hence

$$\int_C \bar{F} \cdot d\bar{r} = \frac{a^3}{3} + (-ab^2) + \left(\frac{a^3}{3} + ab^2\right)$$

$$= -2ab^2$$

Ex: Evaluate $\int_C \bar{F} \cdot d\bar{r}$ where

$$\bar{F} = (3x^2 + 6y)\,\hat{i} - 14yz\,\hat{j} + 20xz^2\,\hat{k}$$

and C is the straight line joining the points (0,0,0) to (1,1,1)

Soln: Let (x,y,z) be the any point on curve C.

∴ Its position vector $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\Rightarrow d\bar{r} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$$

$$\bar{F} \cdot d\bar{r} = (3x^2 + 6y)\,dx - 14yz\,dy + 20xz^2\,dz \quad \text{---(1)}$$

Now The path of integration is a straight line joining points A(0,0,0) and B(1,1,1)

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∴ eqn of line AB

$$\frac{x-0}{0-1} = \frac{y-0}{0-1} = \frac{z-0}{0-1}$$

$$\Rightarrow x = y = z$$

$$\Rightarrow dx = dy = dz$$

Substituting these values in $\bar{F} \cdot d\bar{r}$ and integrating over $x=0$ to $x=1$

$$\therefore \int_C \bar{F} \cdot d\bar{r} = \int_0^1 (3x^2+6y)\,dx - 14yz\,dy + 20xz^2\,dz$$

$$= \int_0^1 (3x^2 + 6x - 14x^2 + 20x^3)\,dx$$

$$= \int_0^1 (20x^3 - 11x^2 + 6x)\,dx$$

$$= 20\frac{x^4}{4} - 11\frac{x^3}{3} + 6\frac{x^2}{2}\Big]_0^1$$

$$= \frac{13}{4}$$

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Ex: Find the workdone by the ~~force~~ Force $\bar{F} = 16y\,\hat{i} + (3x^2+2)\,\hat{j}$ in moving a particle once round of the right half of the ellipse $x^2 + a^2y^2 = a^2$ from $(0,1)$ to $(0,-1)$

Soln:

let $\bar{F} = 16y\,\hat{i} + (3x^2+2)\,\hat{j} + 0\hat{k}$ ——— (1)

let let (x,y,z) be any point on the curve C.

$\therefore \bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$

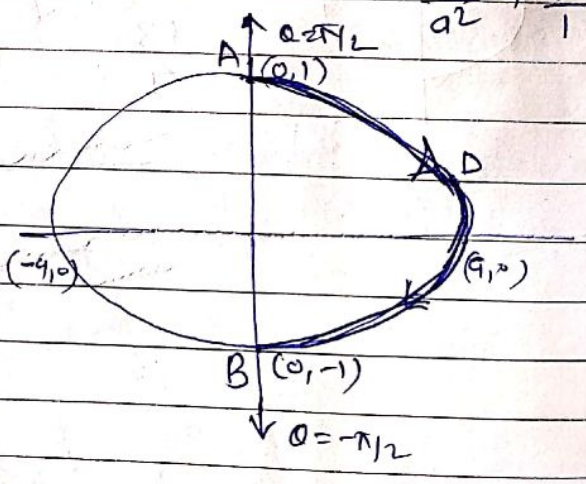
$\Rightarrow d\bar{r} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$

$\therefore \boxed{\bar{F}\cdot d\bar{r} = 16y\,dx + (3x^2+2)\,dy + 0}$ ——— (2)

Since

Note: The path of integration is right half of the ellipse $x^2 + a^2y^2 = a^2$ from pt $(0,1)$ to $(0,-1)$

Given ellipse (curve) is $x^2 + a^2y^2 = a^2 \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{1} = 1$



The parametric eqn. of given curve ellipse are

$x = a\cos\theta$ & $y = 1\sin\theta$

$\Rightarrow dx = -a\sin\theta\,d\theta$

$dy = \cos\theta\,d\theta$

at point $A(0,1) \Rightarrow y = 1 \Rightarrow \sin\theta = 1$.

$\Rightarrow \theta = \sin^{-1}1 = \pi/2$

at point $B(0,-1) \Rightarrow y = -1 \Rightarrow \sin\theta = -1$

$\Rightarrow \theta = \sin^{-1}(-1) = -\pi/2$

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Integrating $\bar{F}.d\bar{r}$ over the curve ADB ie. from $\theta=\pi/2$ to $\theta=-\pi/2$

Thus

$$\text{Work done} = \int_C \bar{F}.d\bar{r}$$

$$= \int_{ADB} 16y\,dx + (3x^2+2)\,dy$$

$$= \int_{\pi/2}^{-\pi/2} 16\sin\theta\,(-a\sin\theta\,d\theta) + (3a^2\cos^2\theta+2)(\cos\theta\,d\theta)$$

$$= -\int_{-\pi/2}^{\pi/2} (-16a\sin^2\theta + 3a^2\cos^3\theta + 2\cos\theta)\,d\theta$$

$$= -2\int_0^{\pi/2} (-16a\sin^2\theta + 3a^2\cos^3\theta + 2\cos\theta)\,d\theta$$

$$= 2\int_0^{\pi/2} 16a\sin^2\theta + (-3a^2)\cdot 2\int_0^{\pi/2}\cos^3\theta + -4\int_0^{\pi/2}\cos\theta\,d\theta$$

$$= 16a\,\beta\left(\tfrac{2+1}{2},\tfrac{0+1}{2}\right) + (-3a^2)\cdot\beta\left(\tfrac{0+1}{2},\tfrac{3+1}{2}\right) - 4\left(\sin\theta\right)_0^{\pi/2}$$

$$= 16a\,\beta(3/2,1/2) - 3a^2\,\beta(1/2,2) - 4(1-0)$$

$$= 16a\,\frac{\Gamma 3/2\;\Gamma 1/2}{\Gamma 2} - 3a^2\,\frac{\Gamma 1/2\cdot\Gamma 2}{\Gamma 1/2+2} - 4$$

$$= 16a\left(\frac{1/2\,\Gamma}{1}\right) - 3a^2\,\frac{\sqrt{\pi}/2}{\Gamma 5/2} - 4$$

$$= 8a\pi - \frac{3a^2\;\Gamma 1/2}{3/2\;1/2\;\Gamma 1/2} - 4$$

$$\text{Work done} = 8a\pi - 4a^2 - 4$$

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H.W

Ex ① Evaluate $\int \bar{F} \cdot d\bar{r}$ from the pt. $A(0,0,0)$ to $B(1,1,1)$ along the curve $\bar{r} = t\hat{i} + t^2\hat{j} + kt^3$ given

$$\bar{F} = xy\,\hat{i} - z^2\hat{j} + xyz\,\hat{k}$$

Ans $1/3$

Hint

let $x = t,\ y = t^2,\ z = t^3 \Rightarrow \boxed{\bar{F} = t^3\hat{i} - t^6\hat{j} + t^6\hat{k}}$

$\therefore d\bar{r} = \dfrac{d\bar{r}}{dt} = \dfrac{dx}{dt}i + \dfrac{dy}{dt}j + \dfrac{dz}{dt}k = i + 2tj + 3t^2k$

$\therefore$ and integrate over $t=0$ to $t=1$.

Ex ② Find the work done in moving particle once round the circle in xy-plane, if the circle has centre at origin and radius 3 and $\bar{F}$ force is given by

$$\bar{F} = (2x - y + z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y + yz)\hat{k}$$

Hint,

Given curve is circle $x^2 + y^2 = 3^2$

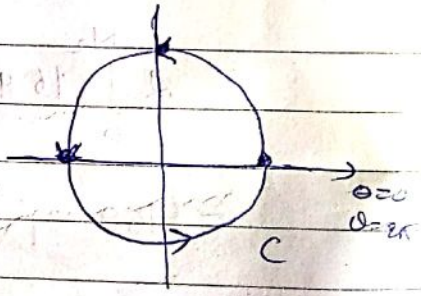
Ans 18π

$\therefore$ parametric eqn is

$x = 3\cos\theta,\quad y = 3\sin\theta \ \&\ z = 0$

$\therefore \bar{r} = 3\cos\theta\, i + 3\sin\theta\, j + 0k$

$\therefore \dfrac{d\bar{r}}{d\theta} = -3\sin\theta\, i + 3\cos\theta\, j$

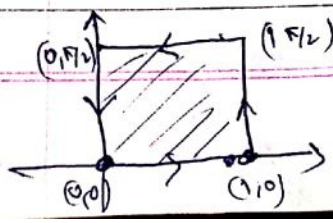


also $\bar{F} = (6\cos\theta - 3\sin\theta + 0)\hat{i} + (3\cos\theta + 3\sin\theta - 0)\hat{j} + (9\cos\theta - 6\sin\theta + 0)\hat{k}$

$\therefore$ Find Work done $= \int_{\theta=0}^{2\pi} \bar{F} \cdot d\bar{r} = 18\pi$.

Ex ③ If $\bar{F} = e^x \sin y\,\hat{i} + e^x \cos y\,\hat{j}$, evaluate $\int_C \bar{F} \cdot d\bar{r}$

where C is rectangle whose vertices are $(0,0)$, $(1,0)$, $(1,\pi/2)$ $(0,\pi/2)$



Ans = 0

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Green's thm :⇒

Statement :

If $M(x,y)$ and $N(x,y)$ and their derivatives $\frac{\partial M}{\partial y}$, $\frac{\partial N}{\partial x}$ are continuous in some region R of xy-plane bounded by a closed curve C. then.

$$\oint_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

(# Direction of line integral should be anticlockwise)

Ex : Verify Green's thm for $\oint_C [(x-y)\,dx + 3xy\,dy]$

where C is the boundary of the region R bounded by parabolas $x^2 = 4y$ and $y^2 = 4x$.

Soln :

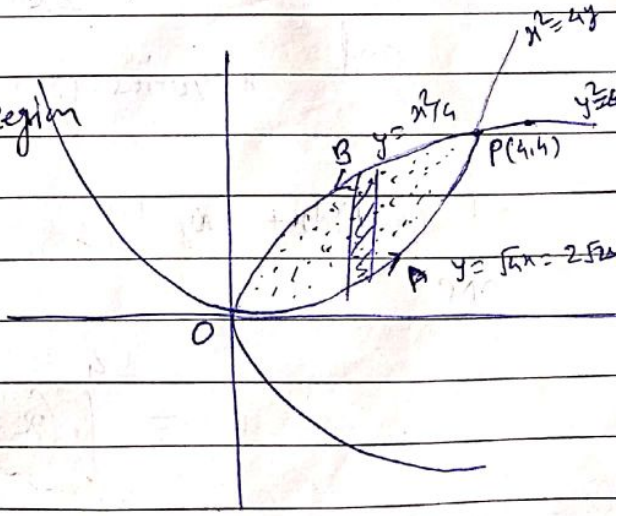
let C is the boundary of Region bounded by parabolas, $y^2 = 4x$ & $x^2 = 4y$

∴ point of intersection is

$$y^2 = 4x$$
$$\Rightarrow \left(\frac{x^2}{4}\right)^2 = 4x$$
$$\Rightarrow x^4 = 64x$$
$$\Rightarrow x(x^3 - 64) = 0 \Rightarrow x = 0, \& x = 4$$
$$\Rightarrow y = 0, y = 4$$

∴ point of intersection is (4, 4)

Thus curve C is a closed curve OAPBO.



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Thus let $M(x,y) = x-y$ $\quad N(x,y) = 3xy$

$\Rightarrow \frac{\partial M}{\partial y} = -1$ $\quad \frac{\partial N}{\partial x} = 3y$

Thus $\Rightarrow \oint_C M dx + N dy =$

$\oint_C [(x-y)dx + 3xy\,dy] = \int_{OAPBO}$

$$\oint_C M dx + N dy = \oint_C (x-y)dx + 3xy\,dy$$

$$= \int_{OAP} (x-y)\,dx + 3xy\,dy + \int_{PBO} (x-y)\,dx + 3xy\,dy \quad \text{---(2)}$$

Along curve OAP

i.e. $x^2 = 4y$

$\Rightarrow 2x\,dx = 4\,dy \quad \Rightarrow dy = \frac{1}{2}x\,dx$

and

x varies from $x=0$ to $x=4$

$$\therefore \int_{OAP} (x-y)dx + 3xy\,dy = \int_{x=0}^{4} \left(x - x^2/4\right) dx + 3x\frac{x^2}{4} \cdot \frac{1}{2}x\,dx$$

$$= \int_0^4 \left(x - x^2/4 + 3/8\, x^3 \cdots\right) dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{12} + \frac{3}{8}\frac{x^5}{5}\right]_0^4$$

$$= \left(8 - \frac{16}{3} + \frac{384}{5}\right) - (0)$$

$$= \frac{1192}{15}$$

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Along PBO : $y^2 = 4x \Rightarrow \frac{1}{4}y^2 = x$

$\therefore \frac{1}{4} 2y\,dy = dx \Rightarrow dx = \frac{1}{2}y\,dy$

$\rightarrow$ Here x varies from $y=4$ to $y=0$

$$\therefore \int_{PBO} (x-y)\,dx + 3xy\,dy = \int_{y=4}^{0} \left(\frac{1}{4}y^2 - y\right)\frac{1}{2}y\,dy + 3\frac{y^2}{4}\cdot y\cdot dy$$

$$= \int_{y=4}^{0} \left(\frac{1}{8}y^3 - \frac{y^2}{2} + \frac{3}{4}y^3\right)dy$$

$$= \frac{y^4}{32} - \frac{y^3}{6} + \frac{3y^4}{16}\Big]_4^0 = \left[\frac{7y^4}{32} - \frac{y^3}{6}\right]_4^0$$

$$= \left[0 - \left(\frac{7(256)}{32} - \frac{64}{6}\right)\right] = \frac{-136}{3}$$

$\therefore$ From (2)

$$\oint_C M\,dx + N\,dy = \frac{1192}{15} - \frac{136}{3} = \frac{512}{15} \quad \text{---(3)}$$

Now,

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)dx\,dy = \iint_R (3y+1)\,dx\,dy$$

In Region R, draw a vertical strip along which y varies from ~~$y=2\sqrt{x}$~~ to $y = x^2/4$ to $y = 2\sqrt{x}$ and x varies from $x=0$ to $x=4$

$$\therefore \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)dx\,dy = \int_{x=0}^{4} \int_{y=x^2/4}^{2\sqrt{x}} (3y+1)\,dy\,dx$$

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$$= \int_0^4 \left[\frac{3y^2}{2} + y \right]_{x^2/4}^{2\sqrt{x}} dx$$

$$= \int_0^4 \left[\left(6x + 2\sqrt{x}\right) - \left(\frac{3}{32}x^4 + \frac{x^2}{4}\right) \right] dx$$

$$= \int_0^4 \left(6x + 2(x)^{1/2} - \frac{3}{32} \cdot x^4 - \frac{1}{4}x^2\right) dx$$

$$= 6\frac{x^2}{2} + 2\frac{x^{3/2}}{3/2} - \frac{3}{32}\frac{x^5}{5} - \frac{1}{4}\frac{x^3}{3} \Big]_0^4$$

$$= \left[3x^2 + \frac{4}{3}x^{3/2} - \frac{3}{160}x^5 + \frac{1}{12}x^3 \right]_0^4$$

$$= 48 + 32/3 - 96/5 - 16/3$$

$$\therefore \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy = \frac{512}{15} \quad \text{---------- (4)}$$

from (3) & (4) we get

$$\oint_C M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

$\therefore$ Hence Greens thm verified.

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Ex: Verify Green's thm for $\oint_C (y-\sin x)\,dx + \cos x\,dy$ where C is the plane triangle enclosed by the line $y=0$, $x=\pi/2$, $y=2x/\pi$.

Soln. let

$$\oint_C M\,dx + N\,dy = \oint_C (y-\sin x)\,dx + \cos x\,dy \quad \text{—— (1)}$$

Here

$M(x,y) = y-\sin x$ $\quad\therefore \frac{\partial M}{\partial y} = 1$

$N(x,y) = \cos x$ $\quad \frac{\partial N}{\partial x} = -\sin x$

Let C be the plane triangle ~~bound~~ enclosed by the lines

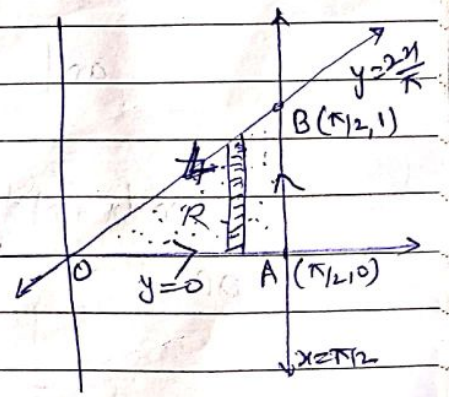
$y=0$ (x-axis). $x=\pi/2$ & $y = 2x/\pi$

$\therefore$ point of intersection of $x=\pi/2$ & $y=2x/\pi$ is $(\pi/2, 1)$

Now line integral

$$\oint_C M\,dx + N\,dy$$

$$= \left[\int_{OA} + \int_{AB} + \int_{BO}\right](M\,dx + N\,dy)$$



—— (2)

Along line OA

$y=0 \Rightarrow dy=0$

and x varies from 0, to $\pi/2$

$$\therefore \int_{OA} M\,dx + N\,dy = \int_{x=0}^{\pi/2} (0-\sin x)\,dx + 0$$

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$$= -\int_0^{\pi/2} \sin x \, dx$$

$$= -\left[-\cos x\right]_0^{\pi/2}$$

$$\int_{OA} M dx + N dy = \left[\cos \pi/2 - \cos 0\right] = 0 - 1 = -1$$

Along AB

$x = \pi/2 \Rightarrow dx = 0$

and y varies from 0 to 1

$$\therefore \int_{AB} M dx + N dy = \int_{y=0}^{1} (y - \sin \pi/2) \cdot 0 + \cos(\pi/2) \cdot dy = 0$$

Along BO

$y = 2x/\pi \Rightarrow dy = 2/\pi \, dx$

and x varies from $x = \pi/2$ to $x = 0$

$$\therefore \int_{AB} M dx + N dy = \int_{x=\pi/2}^{0} \left(\frac{2x}{\pi} - \sin x\right) dx + \cos x \cdot \frac{2}{\pi} dx$$

$$= \frac{x^2}{\pi} + \cos x + \frac{2}{\pi} \sin x \Big]_{\pi/2}^{0}$$

$$= \left[(0 + 1 + 0) - (\pi/4 + 0 + 2/\pi)\right]$$

$$= 1 - \pi/4 + 2/\pi$$

$=$

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Hence (2) $\Rightarrow \oint_C M dx + N dy = -1 + 0 + (1 - \pi/4 - 2/\pi)$

$$= -\left(\frac{\pi^2+8}{4\pi}\right) \quad \text{---------- (3)}$$

Now: Consider a region R as shown in fig.

Draw a vertical strip along which

y varies from $y = 0$ to $y = \frac{2x}{\pi}$

x varies from $x = 0$ to $x = \pi/2$

$$\therefore \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx dy = \int_{x=0}^{\pi/2} \int_{y=0}^{2x/\pi} (-\sin x - 1)\, dy\, dx$$

$$= -\int_{x=0}^{\pi/2} (\sin x + 1) \left[y\right]_0^{2x/\pi} dx$$

$$= -\int_{x=0}^{\pi/2} (\sin x + 1) \left(\frac{2x}{\pi}\right) dx$$

$$= -\frac{2}{\pi} \int_{x=0}^{\pi/2} (x + x \sin x)\, dx$$

$$= \frac{-2}{\pi} \left[\frac{x^2}{2} + x(-\cos x) - (-\sin x)\right]_0^{\pi/2}$$

$$= \frac{-2}{\pi} \left[\left(\frac{\pi^2}{8} + 0 + 1\right) - 0\right]$$

$$= -\frac{(\pi^2+8)}{4\pi} \quad \text{---------- (4)}$$

from (3) & (4)

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx dy$$

$\therefore$ Green's thm is verified.

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Ex: Using Green's thm., Evaluate $\oint_C (-x^2y\,dx + xy^2\,dy)$ where C is the cardioid $r = a(1+\cos\theta)$.

Sol: let $M(x,y) = -x^2y \Rightarrow \frac{\partial M}{\partial y} = -x^2$

$N(x,y) = xy^2 \Rightarrow \frac{\partial N}{\partial x} = y^2$

- By using Green's thm., we have

$$\oint_C -x^2y\,dx + xy^2\,dy = \oint_C M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

$$= \iint_R (y^2 + x^2)\,dx\,dy. \quad \text{---------- (1)}$$

Here R is region bound. enclosed by curve C i.e. cardioid $r = a(1+\cos\theta)$.

∴ By changing into polar form.

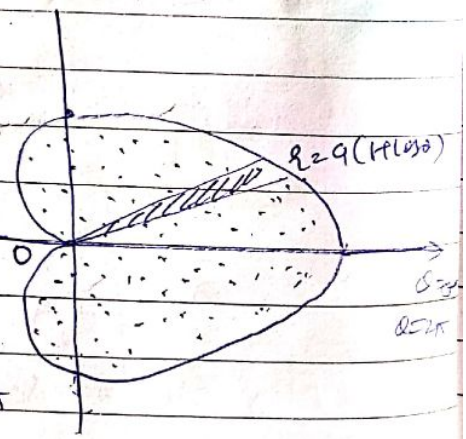
put $x = r\cos\theta$, $y = r\sin\theta$

& $dx\,dy = r\,dr\,d\theta$

Draw a radial strip in Region R along which

r varies from $r = 0$ to $r = a(1+\cos\theta)$

and θ varies from $\theta = 0$ to $\theta = 2\pi$



∴ Thus eqn (1) $\Rightarrow$

$$\oint_C -x^2y\,dx + xy^2\,dy = \iint_R (x^2+y^2)\,dx\,dy$$

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$$= \int_{\theta=0}^{2\pi} \int_{r=0}^{a(1+\cos\theta)} (r^2\cos^2\theta + r^2\sin^2\theta)\cdot r\,dr\,d\theta$$

$$= \int_{\theta=0}^{2\pi} \left[\int_{r=0}^{a(1+\cos\theta)} r^3\,dr \right] d\theta$$

$$= \int_{\theta=0}^{2\pi} \left[r^4/4 \right]_0^{a(1+\cos\theta)} \cdot d\theta$$

$$= \frac{1}{4} \int_{\theta=0}^{2\pi} a^4(1+\cos\theta)^4\,d\theta$$

$$= \frac{a^4}{4} \int_{\theta=0}^{2\pi} (2\cos^2\theta/2)^4\,d\theta$$

$$= 4a^4 \int_{\theta=0}^{2\pi} \cos^8\theta/2\,d\theta$$

put $\theta/2 = t \Rightarrow d\theta = 2dt$

as $\theta = 0 \Rightarrow t = 0$

as $\theta = 2\pi \Rightarrow t = \pi$

$$\therefore \oint_C x^2y\,dx + xy^2\,dy = 4a^4 \cdot \int_0^{\pi} \cos^8 t \cdot 2dt$$

$$= 8a^4 \int_0^{\pi} \cos^8 t \cdot dt$$

$$= 8a^4 \int_0^{2\pi/2} (\cos t)^8 \cdot dt$$

$$= 8a^4 \cdot 2 \int_0^{\pi/2} \cos^8 t\,dt$$

$$\int_0^{2a} f(x)\,dx = 2\int_0^{a} f(x)\,dx$$

if $f(x) = f(2a-x)$

$\therefore (\cos t)^8 = (\cos(\pi - t))^8$

$= (-\cos t)^8 = (\cos\pi\cos t + \sin\pi\sin t)^8$

$= \cos t^8$

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$$= 8a^4 \cdot 2 \int_0^{\pi/2} \sin^0\theta \cdot \cos^8\theta \, d\theta$$

$$= 8a^4 \cdot \beta\left(\frac{0+1}{2}, \frac{8+1}{2}\right)$$

$$= 8a^4 \cdot \frac{\Gamma 1/2 \; \Gamma 9/2}{\Gamma 10/2}$$

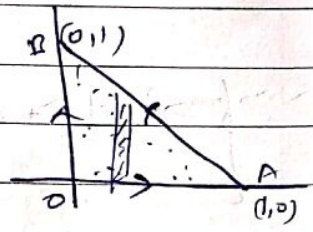
$$= 8a^4 \; \frac{\Gamma 1/2 \cdot 7/2 \cdot 5/2 \cdot 3/2 \cdot 1/2 \; \Gamma 1/2}{4!}$$

$$= \frac{8a^4 \; (7 \times 5 \times 3) \; \pi}{4! \cdot (2 \times 2 \times 2)} = \frac{35 a^4 \pi}{96}$$

Thus $$\boxed{\oint_C (-x^2 y \, dx + xy^2 \, dy) = \frac{35 a^4 \pi}{16}}$$

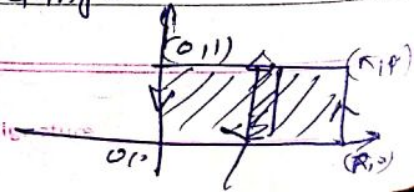
Hw

① Verify Greens thm for $\oint_C (3x^2 - 8y^2)\,dx + (4y - 6xy)\,dy$ where C is the boundary of the region defined by $x=0$, $y=0$ & $x+y=1$



② Evaluate $\oint_C (x^2 - \cosh y)\,dx + (y + \sin x)\,dy$ by using Greens thm where C is the rectangle with vertices $(0,0)$, $(\pi,0)$, $(\pi,1)$, $(0,1)$

Ans) $\pi(\cosh 1 - 1)$



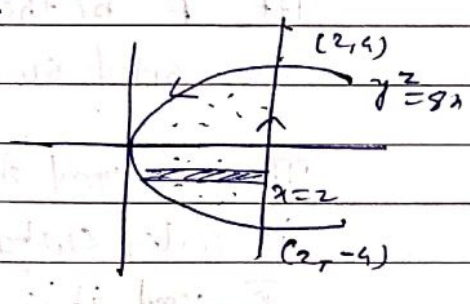
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(3) verify Green's thm in the plane for

$$\oint_C (x^2-2xy)\,dx + (x^2y+3)\,dy$$

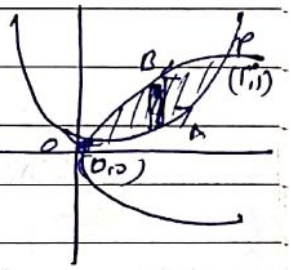
where C is boundary of the region enclosed by $y^2 = 8x$ & $x = 2$



(4) verify Green's thm in the plane for

$$\oint_C (3x^2-8y^2)\,dx + (4y-6xy)\,dy$$

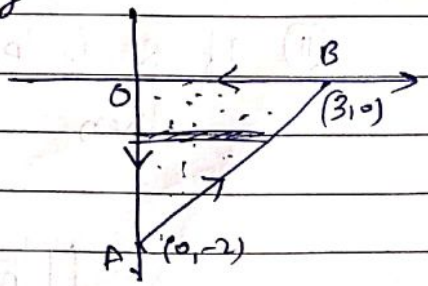
where C is boundary of the region bounded by $y = \sqrt{x}$ and $y = x^2$



(5) Verify Green's thm in the plane for

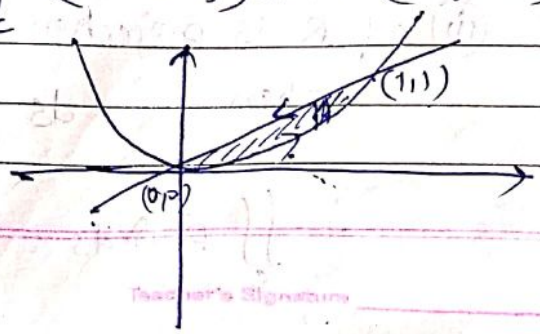
$$\oint_C (3x^2-8y^2)\,dx + (4y-6xy)\,dy$$

where C is the boundary of the region bounded by $x \geq 0$, $y \geq 0$ and $2x-3y=6$



(6) Verify Green's thm for $\oint_C (x^2-2xy)\,dx + (x^2y+3)\,dy$

where C is the boundary of the region bounded by $y = x^2$ and $y = x$

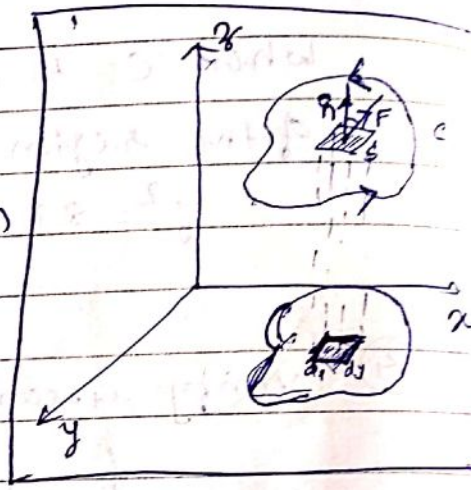


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Surface Integral :⇒

Let $\vec{F}$ be the vector point function defined over a two sided surface S of closed curve C.



Then integral of $\vec{F}$ over surface S is called surface integral of ~~vector function~~ $\vec{F}$ ~~and~~ it is denoted by

$$\iint_S \vec{F}\cdot\hat{n}\,ds \quad \underline{OR} \quad \iint_S \vec{F}\cdot d\vec{s}$$

where $\hat{n}$ is unit vector normal to the surface S.

Note:

(i) If R_1 is projection of surface S in xy-plane (

then ~~$\hat{n}=\hat{k}$~~ $\therefore ds = \dfrac{dx\,dy}{|\hat{n}\cdot\hat{k}|}$ = ~~dxdy~~

$$\therefore \iint_S \vec{F}\cdot d\vec{s} = \iint_S \vec{F}\cdot\hat{n}\,ds = \iint_{R_1} \vec{F}\cdot\hat{n}\,\frac{dx\,dy}{|\hat{n}\cdot\hat{k}|}$$

(ii) If R is projection of surface S in yz plane

then $$ds = \frac{dy\,dz}{|\hat{n}\cdot\hat{i}|}$$

$$\therefore \iint_S \vec{F}\cdot\hat{n}\,ds = \iint_R \vec{F}\cdot\hat{n}\,\frac{dy\,dz}{|\hat{n}\cdot\hat{i}|}$$

(iii) If R is projection of surface S in zx plane

then $$ds = \frac{dx\,dz}{|\hat{n}\cdot\hat{j}|}$$

$$\therefore \iint_S \vec{F}\cdot\hat{n}\,ds = \iint_R \vec{F}\cdot\hat{n}\,\frac{dx\,dz}{|\hat{n}\cdot\hat{j}|}$$

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The rate of flow of fluid, particle or energy.
or
a quantity expressing the strength of a field of force in a given area.

or
the action or process of flowing or flowing out

Flux : (Volume of the fluid flowing out from surface S per unit time)

Let $\vec{F}$ represent velocity of the fluid at any point on a closed surface, S.

Then the surface integral $\iint_S |\vec{F} \cdot \hat{n}| ds$

represents the flux of $\vec{F}$ over S.

Note : If $\iint_S |\vec{F} \cdot \hat{n}| ds = 0$

Then $\vec{F}$ is called solenoidal vector point function.



Note: If S be the surface given by $\phi(x,y,z) = 0$

then $\hat{n}$ = unit normal to surface S $= \dfrac{\nabla\phi}{|\nabla\phi|}$.

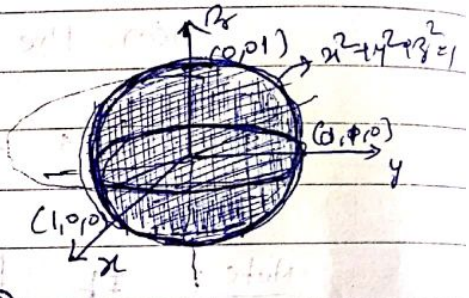


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Ex. Evaluate: $\iint_S \bar{F} \cdot \hat{n}\, ds$ over the surface of the sphere

$x^2+y^2+z^2=1$ in the positive octant.

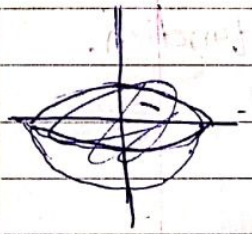
where $\bar{F} = yz\,\hat{i} + xz\,\hat{j} + xy\,\hat{k}$.



Soln: Let

$$\bar{F} = yz\,\hat{i} + xz\,\hat{j} + xy\,\hat{k} \quad \text{---(1)}$$

given surface is sphere. $x^2+y^2+z^2=1$ ---(2)



Let $\phi(x,y,z) = x^2+y^2+z^2-1 = 0$

$$\nabla\phi = \left(\frac{\partial}{\partial x}\,i + \frac{\partial}{\partial y}\,j + \frac{\partial}{\partial z}\,k\right)\cdot(x^2+y^2+z^2-1)$$

$$= 2x\,i + 2y\,j + 2z\,k$$

$$= 2(xi+yj+zk)$$

and $$|\nabla\phi| = \sqrt{(2x)^2+(2y)^2+(2z)^2} = \sqrt{4(x^2+y^2+z^2)}$$

$$= \sqrt{4(1)} \quad \because (2)$$

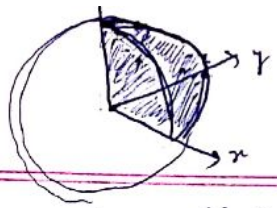
$$= 2$$

Thus unit normal to surface ϕ

ie $$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2(xi+yj+zk)}{2}$$

$$\boxed{\hat{n} = xi+yj+zk} \quad \text{---(3)}$$

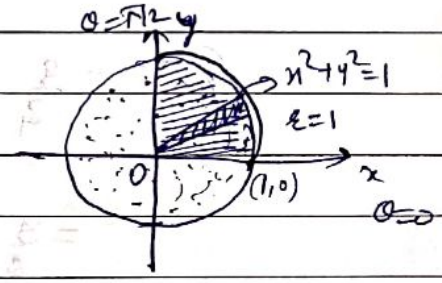
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Now, let R be the region obtained by taking projection of the sphere $x^2+y^2+z^2=1$ on xy-plane (in positive octant)

which Thus R is a region of circle

→ $x^2+y^2=1$ in xy-plane ($z=0$)



also, $ds = \dfrac{dx\,dy}{|\hat{n}\cdot\hat{k}|}$

$= \dfrac{dx\,dy}{|(xi+yj+zk)\cdot\hat{k}|} = \dfrac{dx\,dy}{z}$ ——— (4)

Thus.

$$\iint_S \bar{F}\cdot\hat{n}\,ds = \iint_R (yz\,\hat{i}+xz\,\hat{j}+xy\,\hat{k})\cdot(x\,\hat{i}+y\,\hat{j}+z\,\hat{k})\cdot\frac{dx\,dy}{z}$$

$$= \iint_R 3xyz\cdot\frac{dx\,dy}{z} = 3\iint_R xy\cdot dx\,dy \quad \text{——— (5)}$$

Now.

In Region R draw a radial strip.

where r varies from $r=0$ to $r=1$

θ varies from $\theta=0$ to $\theta=\pi/2$

$$(5) \Rightarrow \iint_S \bar{F}\cdot\hat{n}\,ds = 3\int_{\theta=0}^{\pi/2}\int_{r=0}^{1} (r\cos\theta)\cdot(r\sin\theta)\cdot r\,dr\,d\theta$$

$$= 3\int_{\theta=0}^{\pi/2}\int_{r=0}^{1} r^3\cos\theta\sin\theta\,dr\,d\theta$$

$$= 3\int_{\theta=0}^{\pi/2} \cos\theta\cdot\sin\theta\cdot\left[\frac{r^4}{4}\right]_0^1 d\theta$$

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$$= 3 \int_{\theta=0}^{\pi/2} \cos\theta \cdot \sin\theta \left[\frac{1}{4} - 0\right] d\theta$$

$$= \frac{3}{4} \int_{\theta=0}^{\pi/2} \left[\frac{\sin 2\theta}{2}\right] d\theta$$

$$= \frac{3}{8} \left[\frac{-\cos 2\theta}{2}\right]_0^{\pi/2}$$

$$= \frac{-3}{16} [\cos\pi - \cos 0] = \frac{-3}{16}[-1-1]$$

$$= \frac{6}{16} = \frac{3}{8}$$

$$\therefore \boxed{\iint_S \bar{F} \cdot \hat{n}\, ds = 3/8}$$

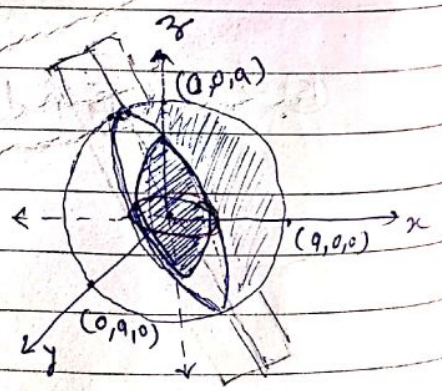


Ex: Find the flux of $\vec{F} = \hat{i} - \hat{j} + xyz\hat{k}$ through the circular region S obtained by cutting sphere $x^2+y^2+z^2 = a^2$ with plane $y=x$

Soln:

let $\vec{F} = \hat{i} - \hat{j} + xyz\,\hat{k}$ — (1)

let S be the surface obtained in x-z plane by intersection of sphere $x^2+y^2+z^2=a^2$ with the plane $y=x$



$\therefore$ i.e. $2x^2 + z^2 = a^2$

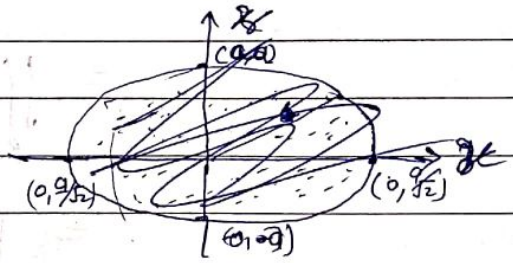
$$\Rightarrow \frac{x^2}{a^2/2} + \frac{z^2}{a^2} = \frac{1}{2} \qquad \frac{x^2}{(a/\sqrt{2})^2} + \frac{z^2}{a^2} = 1$$

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$\Rightarrow \frac{x^2}{a^2} + \frac{z^2}{(\sqrt{2}a)^2} = \left(\frac{a}{\sqrt{2}}\right)^2$ is an ellipse

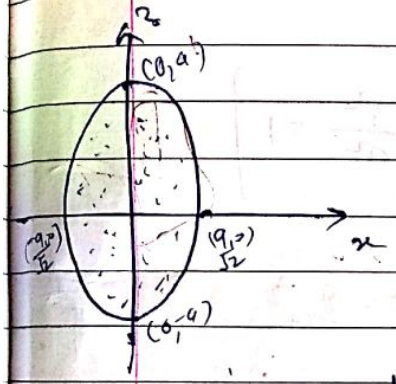
is an ellipse in xz-plane



Thus surface $S\phi$ is an ellipse

$\Rightarrow 2x^2 + z^2 = a^2$

Now normal to the surface S is also normal to plane $y = x$



$\therefore$ let $\phi = y - x = 0$

$\therefore \nabla\phi = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right) \cdot (y - x)$

$= -\hat{i} + \hat{j}$

& $|\nabla\phi| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\therefore$ normal vector $\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{1}{\sqrt{2}}(-\hat{i} + \hat{j})$

Now let R be the projection of the surface S on xz-plane which is an ellipse $2x^2 + y^2 = a^2$

$\therefore ds = \frac{dx\,dz}{|\hat{n} \cdot \hat{j}|} = \frac{dx\,dz}{\left|\left(\frac{-1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}\right) \cdot \hat{j}\right|}$

$= \frac{dx\,dz}{\left|\frac{1}{\sqrt{2}}\right|} = \sqrt{2}\,dx\,dz$

Now.

$\text{flux} = \iint_S |\vec{F} \cdot \hat{n}|\,ds$

$= \iint_R \left|(0 - j + xyz\hat{k}) \cdot \frac{1}{\sqrt{2}}(\hat{i} + \hat{j})\right| \cdot \sqrt{2}\,dx\,dy$

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$$\text{flux} = \iint_R \left|(i - j + xyz\,k)\cdot(-i + j)\right| \cdot dx\,dz$$

$$= \iint_R |-1-1| \; dx\,dz$$

$$= |-2| \iint_R dz\,dx$$

$$= 2\left[\iint_R dz\,dx\right]$$

$$= 2\left[\text{Area of ellipse } \frac{x^2}{(a/\sqrt{2})^2} + \frac{z^2}{a^2} = 1\right]$$

$$= 2 \cdot \pi \cdot \left(\frac{a}{\sqrt{2}}\right) \cdot a$$

$$\boxed{\text{flux} = \sqrt{2}\pi a^2}$$

$x^2 = \frac{a^2 - z^2}{2}$

$x = \pm\sqrt{\frac{a^2 - z^2}{2}}$

$z = -a \;\; +a$

Note: ① area of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\boxed{A = \pi \cdot a \cdot b}$$

~~Area of a circle~~ $x^2 + y^2 = R^2$

$$\Rightarrow \frac{x^2}{R^2} + \frac{y^2}{R^2} = 1$$

$$A = \pi R \cdot R$$

$$= \pi R^2$$

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★ Stoke's Thm :

Statement : If S be an open surface bounded by a closed curve C and $\vec{F}$ be a continuous and differentiable vector function, then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, ds$$

where $\hat{n}$ is unit normal (outward) at any pt of the surface S.

⟺

Ex : Verify stokes thm for $\vec{F} = (x+y)\hat{i} + (y+z)\hat{j} - x\hat{k}$ and S is the surface of the plane $2x+y+z=2$ in first octant.

Soln : let

$$\vec{F} = (x+y)\hat{i} + (y+z)\hat{j} - x\hat{k} \quad \text{— (1)}$$

$$\therefore \nabla \times \vec{F} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x+y & y+z & -x \end{vmatrix}$$

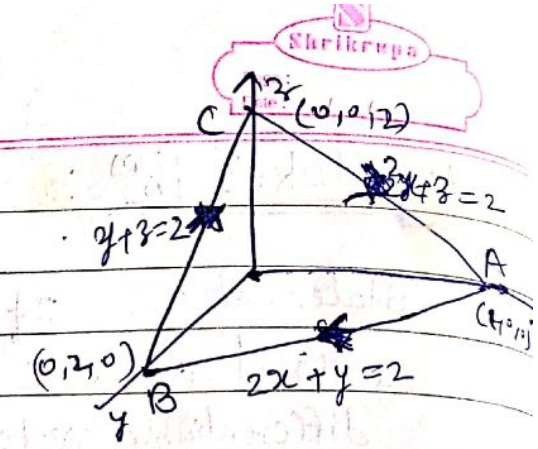
$$= i(0-1) + j(-1-0) + k(0-1)$$

$$\nabla \times \vec{F} = -\hat{i} + \hat{j} - \hat{k} \quad \text{— (2)}$$

let given surface $\phi(x,y,z) = 2x+y+z-2 = 0$

$$\therefore \nabla\phi = \left(\frac{\partial}{\partial x}i + \frac{\partial}{\partial y}j + \frac{\partial}{\partial z}k\right) \cdot (2x+y+z-2)$$

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Flux = $\iint_R (-1 \cdot -1)\, dx\, dz$

$= 2 \iint_R dx\, dz$

$= 2$

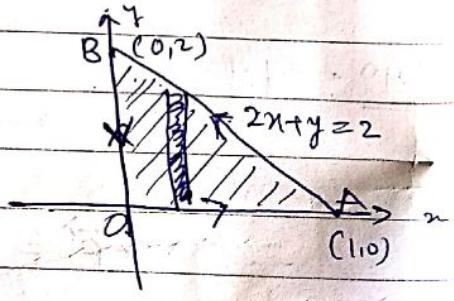
$\Rightarrow \nabla\phi = 2\hat{i} + \hat{j} + \hat{k}$

$\therefore |\nabla\phi| = \sqrt{2^2+1^2+1^2} = \sqrt{6}$

Hence

$\hat{n}$ = unit normal to the surface $= \frac{\nabla\phi}{|\nabla\phi|} = \frac{1}{\sqrt{6}}(2i+j+k)$

Now: let R be the projection of plane $2x+y+z=2$ on xy-plane ($z=0$) in first octant. is ΔOAB.



$\therefore ds = \frac{dx\,dy}{|\hat{n}\cdot\hat{k}|} =$

$= \frac{dx\,dy}{\left|\frac{1}{\sqrt{6}}(2i+j+k)\cdot\hat{k}\right|}$

$= \frac{dx\,dy}{\left|\frac{1}{\sqrt{6}}\right|} = \sqrt{6}\, dx\, dy$

Thus

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \cdot ds = \iint_R (\nabla \times \vec{F}) \cdot \hat{n} \left(\sqrt{6}\, dx\, dy\right)$$

$$= \iint_R (-i+j-k) \cdot \frac{1}{\sqrt{6}} (2i+j+k) \cdot \sqrt{6}\, dx\, dy$$

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$$\iint_S (\nabla \times \bar{F}) \cdot \hat{n}\, ds = \iint_R (-2+1-1)\, dx\, dy$$

$$= \iint_R -2\, dx\, dy \qquad \text{— (3)}$$

Here Region R is bounded by ~~in~~ $x=0$, $y=0$ & line $2x+y=2$

∴ Draw a vertical strip along which

y varies from $y=0$ to $y = 2-2x = 2(1-x)$

x varies from $x=0$ to $x=1$

$$\therefore (3) \Rightarrow \iint_S (\nabla \times \bar{F}) \cdot \hat{n}\, ds = -2 \int_{x=0}^{1} \int_{y=0}^{2-2x} dy\, dx$$

$$= -2 \int_{x=0}^{1} \left[y \right]_0^{2-2x} dx$$

$$= -2 \int_{x=0}^{1} (2-2x)\, dx$$

$$= -2 \left[2x - x^2 \right]_0^1 = -2(2-1) = -2$$

$$\therefore \iint_S (\nabla \times \bar{F})\, \hat{n} \cdot ds = -2 \qquad \text{— (4)}$$

OR from (3)

$= \iint_R -2\, dx\, dy$

$= -2 \iint_R dx\, dy$

$= -2$ [Area of ΔOAB]

$= -2 \left[\frac{1}{2} \times \text{base} \times \text{height}\right]$

$= -2 \left[\frac{1}{2} \cdot (1) \cdot (2)\right]$

$= -2$

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To evaluate $\oint_C \bar{F} \cdot d\bar{r}$.

Here. $\bar{F} \cdot d\bar{r} = [(x+y)\hat{i} + (y+z)\hat{k} - x\hat{n}] \cdot [dx\,i + dy\,j + dz\,k]$

$= (x+y)\,dx + (y+z)\,dy + (-x)\,dz$

also: C is the boundary of region R ie Δ ABC

$\therefore$ By line integral

$$\oint_C \bar{F} \cdot d\bar{r} = \int_{AB} F \cdot dr + \int_{BC} \bar{F} \cdot dr + \int_{CA} \bar{F} \cdot dr \quad \text{——— (5)}$$

In xy-plane

Along AB:

$z = 0 \Rightarrow dz = 0$ & $y = 2 - 2x$

$\Rightarrow dy = -2dx$

and x varies from $x = 1$ to $x = 0$

$$\therefore \int_{AB} \bar{F} \cdot dr = \int_{x=1}^{0} (x + 2 - 2x)\,dx + (x - 2x + 0)\,(-2dx) + 0$$

$$= \int_{x=1}^{0} (x + 2 - 2x + 2 + 4x)\,dx = \int_{x=1}^{0} (3x - 3x)\,dx$$

$$= \left[\frac{3x^2}{2} - 2x\right]_1^0$$

$$= \left[(0) - \left(\frac{3}{2} - 2\right)\right] = -\frac{3}{2} + 2 = \frac{1}{2}$$

In YZ plane: Along line BC

$x = 0$ & $y = 2 - z$

$\Rightarrow dx = 0$ & $dy = -dz$

Here. z varies from $z = 0$, to $z = 2$.

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$$\therefore \int_{BC} \bar{F}\cdot d\bar{r} = \int_{z=0}^{2} [(0+2-z)0 + (2-z+z)(-dz) + (0)\cdot dz]$$

$$= \int_{z=0}^{2} -2\,dz \qquad = -2\left[\frac{z}{}\right]_0^2$$

$$= -4$$

In xz plane : Along line CA

$$y=0 \quad \& \quad 2x+z=2 \Rightarrow z = 2-2x$$

$$\therefore dy = 0 \quad \& \quad dz = -2\,dx$$

Here x varies from $x=0$ to $x=1$

$$\therefore \int_{CA} \bar{F}\cdot d\bar{r} = \int_{x=0}^{1} (x+0)\,dx + (0+2-2x)0 + (-x)(-2\,dx)$$

$$= \int_{x=0}^{1} x\,dx + 2x\,dx \qquad = \int_{x=0}^{1} 3x\,dx$$

$$= 3\left[\frac{x^2}{2}\right]_0^1 \Rightarrow 3/2$$

Hence

eqn (5) $\Rightarrow$ $\oint_C \bar{F}\cdot d\bar{r} = 1/2 - 4 + 3/2 = \frac{-4}{2} = -2$ ——(6)

Now from eqn (4) & (6)

$$\oint_C \bar{F}\cdot d\bar{r} = \iint_S \nabla\times\bar{F}\cdot\hat{n}\,ds$$ is verified

$\therefore$ Stokes thm is verified,

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Ex: Verify stokes thm for $\bar{F} = (x^2+y-4)\hat{i} + 3xy\,\hat{j} + (2xz+z^2)\hat{k}$ over the surface of sphere $x^2+y^2+z^2=16$ above xy-plane.

Soln: To verify stokes thm, we have to show that

$$\oint_C \bar{F}\cdot d\bar{r} = \iint_S (\nabla \times \bar{F})\cdot \hat{n}\, ds$$

let $\bar{F} = (x^2+y-4)\hat{i} + 3xy\,\hat{j} + (2xz+z^2)\hat{k}$ —①

$$\therefore \nabla \times \bar{F} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2+y-4 & 3xy & 2xz+z^2 \end{vmatrix}$$

$$= i(0-0) - j(2z-0) + k(3y-1)$$

$$\boxed{\nabla \times F = 0\,\hat{i} - 2z\,\hat{j} + (3y-1)\,\hat{k}} \quad —②$$

Let given surface is sphere $x^2+y^2+z^2=16$

$$\therefore \phi = x^2+y^2+z^2-16 = 0$$

$$\nabla\phi = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\cdot(x^2+y^2+z^2-16)$$

$$= 2x\,\hat{i} + 2y\,\hat{j} + 2z\,\hat{k}$$

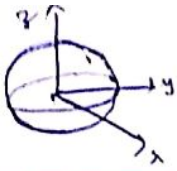
$$\therefore |\nabla\phi| = \sqrt{(2x)^2+(2y)^2+(2z)^2} = \sqrt{4(x^2+y^2+z^2)}$$

$$= \sqrt{4(16)} = \sqrt{64}$$

$$|\nabla\phi| = 8$$

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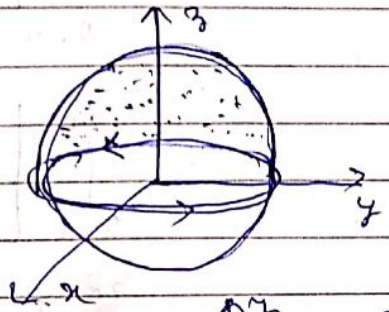
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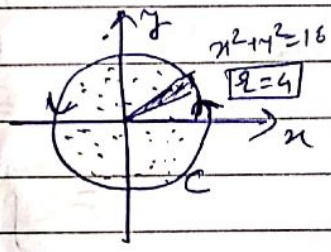
Thus unit normal to the surface ϕ is

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{1}{8}(2x\hat{i}+2y\hat{j}+2z\hat{k})$$

$$\boxed{\hat{n} = \frac{1}{4}(x\hat{i}+y\hat{j}+z\hat{k})}.$$



let R be Projection of sphere $x^2+y^2+z^2=16$ in xy-plane, $(z=0)$.
which is a circle $x^2+y^2=16$



also. $ds = \dfrac{dx\,dy}{|\hat{n}\cdot\hat{k}|}$

$$= \frac{dx\,dy}{\left|\frac{1}{4}(x\hat{i}+y\hat{j}+z\hat{k})\cdot\hat{k}\right|} = \frac{dx\,dy}{\left|\frac{z}{4}\right|}$$

$$\boxed{ds = \frac{4}{z}\,dx\,dy}$$

also in Region R, circle $x^2+y^2=16$ in polar form $r=4$

put $x=r\cos\theta$, $y=r\sin\theta$.

$\Rightarrow dx\,dy = r\,dr\,d\theta$

Here r varies from $r=0$ to $r=4$

θ varies from $\theta=0$ to $\theta=2\pi$

$$\therefore \iint_S (\nabla\times\bar{F})\cdot\hat{n}\cdot ds = \iint_R \left[0\hat{i}-2z\hat{j}+(3y-1)\hat{k}\right]\cdot\frac{1}{4}(x\hat{i}+y\hat{j}+z\hat{k})\cdot\frac{4}{z}\,dx\,dy$$

$$= \iint_R \left(-2zy+(3y-1)z\right)\cdot\frac{dx\,dy}{z}$$

$$= \iint_R (-2y+3y-1)\,dx\,dy$$

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$$\Rightarrow \iint_S \nabla \times F \cdot \hat{n}\, ds = \iint_R (y-1)\, dx\, dy$$

Changing into polar form

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^{4} (r\sin\theta - 1) \cdot r\, dr\, d\theta$$

$$= \int_{\theta=0}^{2\pi} \left[\int_{r=0}^{4} (r^2 \sin\theta - r)\, dr \right] d\theta$$

$$= \int_{\theta=0}^{2\pi} \left[\sin\theta \cdot \left(\frac{r^3}{3}\right) - \frac{r^2}{2} \right]_0^4 d\theta$$

$$= \int_{\theta=0}^{2\pi} \left(\frac{64}{3}\sin\theta - \frac{16}{2} \right) d\theta$$

$$= \left[\frac{64}{3}(-\cos\theta) - \frac{16}{2}\cdot\theta \right]_0^{2\pi}$$

$$= \left[\left(-\frac{64}{3} - 16\pi\right) - \left(-\frac{64}{3} - 0\right) \right] = -16\pi.$$

$$\therefore \boxed{\iint_S \nabla \times \bar{F} \cdot \hat{n}\, ds = -16\pi} \quad \text{---(3)}$$

Now to evaluate. $\oint_C \bar{F}\, d\bar{r}$.

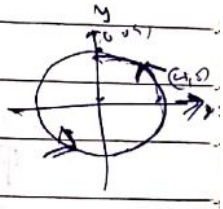
let $\bar{F} \cdot d\bar{r} = (x^2 + y - 4)\, dx + 3xy\, dy + (2x + z^2)\, dz$ ---(4)

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Let C be the boundary of Region R ~~hemisphere~~ S in xy-plane ($z=0$) which is circumference of circle $x^2+y^2=16$.

In xy-plane. $z=0 \Rightarrow dz=0$

$$\therefore \oint_C F \cdot dr = \oint_C \left[(x^2+y-4)\,dx + 3xy\,dy + 0\right]$$



The parametric eqn of circle $x^2+y^2=16$ is

$$x = 4\cos\theta, \quad y = 4\sin\theta$$

$$\Rightarrow dx = -4\sin\theta\,d\theta, \quad dy = 4\cos\theta\,d\theta$$

Along curve C: (i.e. circle).

For the complete circle θ varies from $\theta=0$ to $\theta=2\pi$

and ~~r varies from r=0 to r=4~~

$$\therefore \oint_C \bar{F}\,d\bar{r} = \int_{\theta=0}^{2\pi} \left((4\cos\theta)^2 + 4\sin\theta - 4\right)(-4\sin\theta\,d\theta) + 3(4\cos\theta)(4\sin\theta)\,4\cos\theta\,d\theta$$

$$= \int_{\theta=0}^{2\pi} \left[(16\cos^2\theta + 4\sin\theta - 4)(-4\sin\theta) + 192\cos^2\theta\sin\theta\right] d\theta$$

$$= \int_{\theta=0}^{2\pi} \left[-64\cos^2\theta\sin\theta - 16\sin^2\theta + 16\sin\theta + 192\cos^2\theta\sin\theta\right] d\theta$$

$$= \int_{\theta=0}^{2\pi} (-128\cos^2\theta\sin\theta + 16\sin\theta)\,d\theta + \int_{\theta=0}^{2\pi} 16\sin^2\theta\,d\theta$$

$$= 0 + \int_{\theta=0}^{2\pi} 16\sin^2\theta\,d\theta$$

$$= -16\int_{\theta=0}^{2\pi} \sin^2\theta\,d\theta$$

$\because \int_0^{2a} f(x)\,dx = 0$ if $f(2a-x) = -f(x)$

and $\int_0^{2a} f(x)\,dx = 2\int_0^{a} f(x)\,dx$ if $f(2a-x) = f(x)$

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$\cos 2\theta = 1 - 2\sin^2\theta$

$\Rightarrow \sin^2\theta = \frac{1+}{}$

$$= -16 \int_0^{2\pi} \left(\frac{1+\cos 2\theta}{2}\right) d\theta$$

$$= -8 \left[\theta + \left(\frac{\sin 2\theta}{2}\right)\right]_0^{2\pi}$$

$$= -8 \left[(2\pi + 0) - 0\right]$$

$$\oint_C \bar{F} \cdot d\bar{r} = -16\pi \quad \text{——— (4)}$$

$$= -16(2) \int_0^{\pi} \sin^2\theta \, d\theta$$

$$= -32 (2) \int_0^{\pi/2} \sin^2\theta \, d\theta$$

$$= -32 \, \beta\left(\frac{2+1}{2}, \frac{1}{2}\right)$$

$$= -32 \, \frac{\overline{3/2}\,\overline{1/2}}{\overline{3/2+1/2}}$$

$$= -32 \cdot \frac{1}{2} \cdot \pi$$

$$= -16\pi$$

∴ From eqn (3) & (4)

$$\iint_S \nabla \times \bar{F} \cdot \hat{n} \, ds = \oint_C \bar{F} \cdot d\bar{r}$$

Hence Stokes thm is verified.

Ex: Evaluate $\oint_C (2x-y)\,dx - yz^2\,dy - y^2z\,dz$ using Stokes thm where C is the circle $x^2+y^2=1$ corresponding to the surface of sphere of unit radius.

Soln: let

$$\oint_C (2x-y)dx + yz^2\,dy + y^2z\,dz = \oint_C \bar{F} \cdot d\bar{r} \quad \text{——— (3)}$$

(By Stokes thm.)

$$= \iint_S (\nabla \times F) \cdot \hat{n} \cdot ds. \quad \text{——— (1)}$$

Here

$$\bar{F} \cdot d\bar{r} = \left((2x-y)\hat{i} - yz^2\hat{j} - y^2z\,\hat{k}\right) \cdot (dx\,i + dy\,j + dz\,k)$$

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$\Rightarrow \vec{F} = (2x-y)\,i - yz^2 j - y^2 z\,k$

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (2x-y) & (-yz^2) & (-y^2 z) \end{vmatrix}$$

$= i[-2yz + 2yz] - j[0-0] + k[0-(-1)]$

$= 0i + 0j + k = \hat{k}$

Given surface is unit sphere ie $\phi = x^2+y^2+z^2-1 = 0$

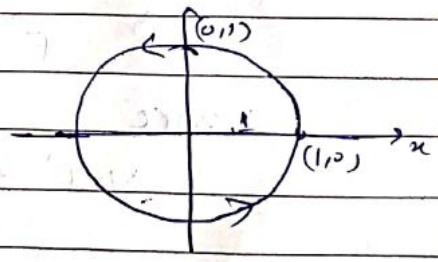
$\therefore \nabla\phi = 2x\,i + 2y\,j + 2z\,k$

& $|\nabla\phi| = \sqrt{4(x^2+y^2+z^2)} = \sqrt{4} = 2$

$\therefore$ Unit normal to the surface ϕ is

$\therefore \hat{n} = \dfrac{\nabla\phi}{|\nabla\phi|} = \dfrac{2(xi+yj+zk)}{2} = xi+yj+zk$

also: C be the circle $x^2+y^2=1$ which is projection of sphere in xy-plane (ie $z=0$)



Thus $ds = \dfrac{dx\,dy}{|\hat{n}\cdot\hat{k}|}$

$= \dfrac{dx\,dy}{|z|}$

From eqn (1)

$$\int_C (2x-y)\,dx - yz^2\,dy - y^2 z\,dz = \iint_S \hat{k}\cdot(xi+yj+zk)\cdot\frac{dx\,dy}{z}$$

$$= \iint_R \frac{z\cdot dx\,dy}{z} = \iint_R dx\,dy$$

= Area of region R = Area of a unit circle $x^2+y^2=1$

$= \pi(1)^2 = \pi$

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Ex: Apply stoks thm to evaluate $\oint_C (x+y)\,dx + (2x-z)\,dy + (y+z)\,dz$

where C is the boundary of the triangle with vertices $(2,0,0)$, $(0,3,0)$ and $(0,0,6)$

Soln: ⇒ By stokes thm, we have

$$\oint_C (x+y)\,dx + (2x-z)\,dy + (y+z)\,dz = \oint_C \vec{F}\cdot d\vec{r}$$

$$= \iint_S (\nabla \times \vec{F})\,\hat{n}\cdot ds \qquad \text{—(1)}$$

Here

$$\vec{F}\cdot d\vec{r} = (x+y)\,dx + (2x-z)\,dy + (y+z)\,dz$$

$$= [(x+y)\,i + (2x-z)\,j + (y+z)\,k]\cdot(dx\,i + dy\,j + dz\,k)$$

$$= \left((x+y)\,i + (2x-z)\,j + (y+z)\,k\right)\cdot d\vec{r}$$

$$\Rightarrow \vec{F} = (x+y)\,\hat{i} + (2x-z)\,\hat{j} + (y+z)\,\hat{k}$$

Hence

$$\nabla \times \vec{F} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (x+y) & (2x-z) & (y+z) \end{vmatrix}$$

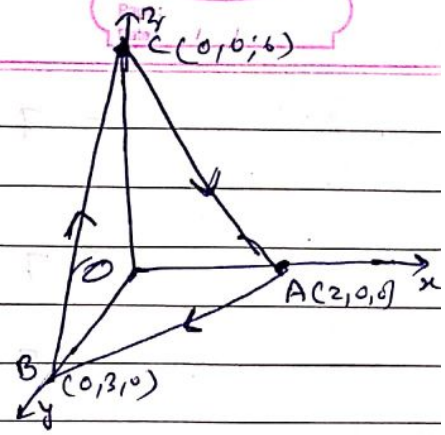
$$= i\,(1+1) - j\,(0-0) + k\,(2-1)$$

$$\boxed{\nabla \times F = 2i + 0j + k}$$

Also C be the boundry of the triangle with vertices $A(2,0,0)$, $B(0,3,0)$, $C(0,0,6)$

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Shrikrupa



Thus S is a surface of plane

$$\frac{x}{2}+\frac{y}{3}+\frac{z}{6}=1$$

$$\Rightarrow \boxed{3x+2y+z=6}$$

Thus consider

$$\phi = 3x+2y+z-6=0$$

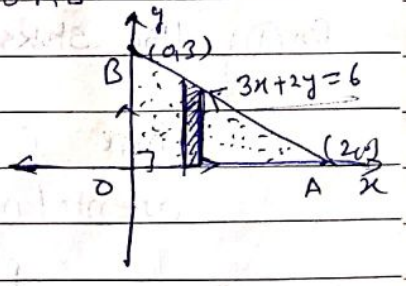
$$\Rightarrow \nabla\phi = \left(\frac{\partial}{\partial x}i+\frac{\partial}{\partial y}j+\frac{\partial}{\partial z}k\right)\cdot(3x+2y+z-6)$$

$$= 3i+2j+k$$

$$\& \quad |\nabla\phi| = \sqrt{3^2+2^2+1^2} = \sqrt{14}$$

Thus unit normal to surface ϕ is $\boxed{\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{1}{\sqrt{14}}(3i+2j+k)}$

Now. let R be the projection on x-y-plane ($z=0$) of a given plane. this is a triangle OAB



$$\therefore ds = \frac{dx\,dy}{|\hat{n}\cdot\hat{k}|} = \frac{dx\,dy}{\left|\frac{1}{\sqrt{14}}(3i+2j+k)\cdot\hat{k}\right|}$$

$$= \frac{dx\,dy}{|1/\sqrt{14}|} = \sqrt{14}\cdot dx\,dy$$

From eq^n (1),

$$\int_C (x+y)\,dx+(2x-z)\,dy+(y+z)\,dz = \iint_S \nabla\times F\cdot\hat{n}\,ds$$

$$= \iint_R (2i+0j+k)\cdot\frac{1}{\sqrt{14}}(3i+2j+k)\cdot\sqrt{14}\,dx\,dy$$

$$= \iint_R (6+1)\cdot dx\,dy$$

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$$= 7 \iint_R dx\,dy$$

$$= 7 \quad [\text{Area of } \Delta OAB]$$

$$= 7 \left(\tfrac{1}{2} \text{ hight} \times \text{base}\right)$$

$$= 7/2 \,(3 \times 2)$$

$$= 21$$

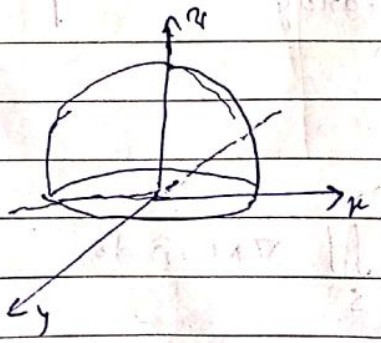
$$\therefore \boxed{\int_C (x+y)\,dx + (2x-z)\,dy + (y+z)\,dz = 21}$$

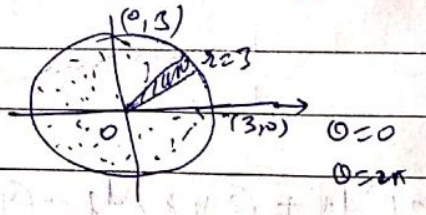
Note:

Ex ① Use stoks thm to evaluate $\oint_C y^2\,dx + xy\,dy + xz\,dz$

where C is curve bounded by hemisphere $x^2+y^2+z^2=9$, $z>0$ oriented in positive direction.

Ans : 0





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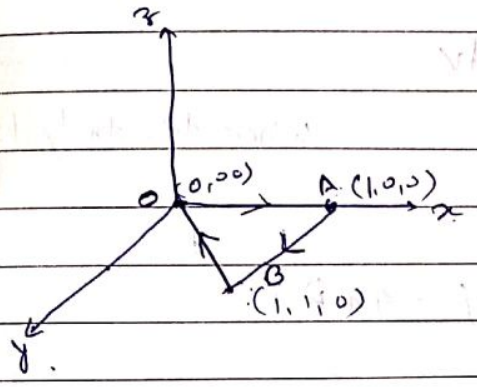
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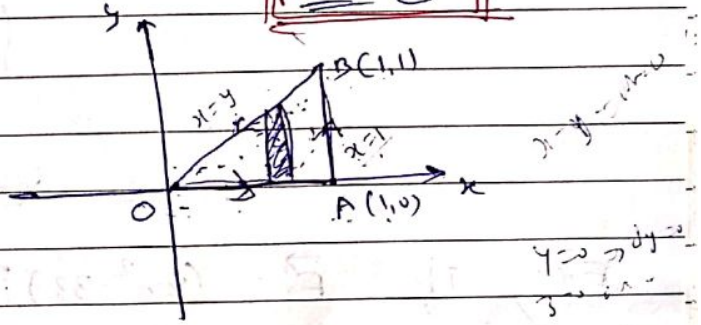
② Evaluate $\oint_C \bar{F} \cdot d\bar{r}$ by using stoks thm

where $\bar{F} = y^2 i + x^2 j - (x+z) k$

and C is the boundary of triangle with vertices at (0,0,0), (1,0,0) and (1,1,0).

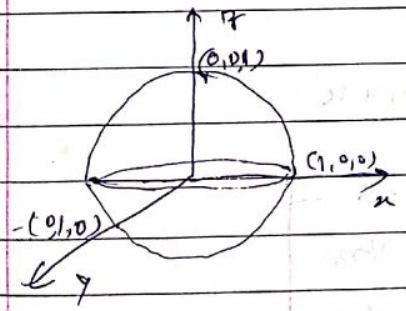


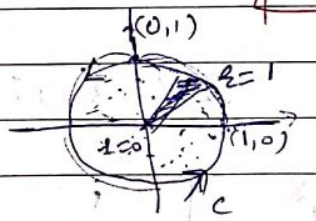
Ans ③



③ Verify Stokes thm for the vector field $\vec{F} = (2x-y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$ over the upper half of the surface $x^2+y^2+z^2=1$ bounded by its projection on xy-plane.

Ans (π)





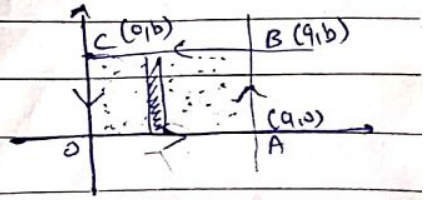
θ = 0 to 2π

④ verify stoks thm for vector field $\bar{F} = (x^2-y^2)\,i + 2xy\,\hat{j}$ in the rectangle in xy-plane bounded by line $x=0$, $x=a$, $y=0$, $y=b$

Ans $2ab^2$

Here: z axis is ~~parallel~~ normal to the rectangle

ie. $\hat{n} = k$



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✱ Volume Integral :

Let $\vec{F}$ be a vector point function and V be the volume enclosed by closed surface S. then the volume integral is given by

$$\iiint_V \vec{F} \cdot dV$$

where $dV = dx\,dy\,dz$.

Ex: If $\vec{F} = (2x^2 - 3z)\,\hat{i} - 2xy\,\hat{j} - 4x\,\hat{k}$

Then evaluate $\iiint_V \nabla \times \vec{F}\, dV$

where V is the closed region bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $2x + 2y + z = 4$

Soln: Let

$$\vec{F} = (2x^2 - 3z)\,i - 2xy\,j - 4x\,k$$

$$\therefore \nabla \times F = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2x^2 - 3z & -2xy & -4x \end{vmatrix}$$

$$= i\left[\frac{\partial}{\partial y}(-4x) - \frac{\partial}{\partial z}(-2xy)\right] - j\left[\frac{\partial}{\partial x}(-4x) - \frac{\partial}{\partial y}(2x^2 - 3z)\right]$$

$$+ k\left[\frac{\partial}{\partial x}(-2xy) - \frac{\partial}{\partial y}(2x^2 - 3z)\right]$$

$$= i[0 - 0] - j[-4 - (-3)] + k[-2y - 0]$$

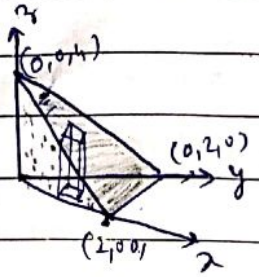
$$= 0i + j - 2yk$$

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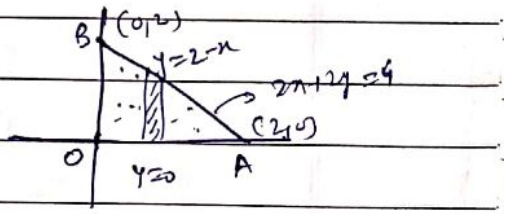
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$\therefore \nabla \times \vec{F} = \hat{j} - 2y\hat{k}$

Let V be the volume bounded by $x=0, y=0, z=0$ & plane $2x+2y+z=4$



projection in xy-plane



Thus z varies from $z=0$ to $z=4-2x-2y$

y varies from $y=0$ to $y=2-x$

x varies from $x=0$ to $x=2$

$$\iiint_V \nabla \times \vec{F} \, dV = \int_{x=0}^{2} \int_{y=0}^{2-x} \int_{z=0}^{4-2x-2y} (\hat{j} - 2y\hat{k}) \, dz \, dy \, dx$$

$$= \int_{x=0}^{2} \int_{y=0}^{2-x} (\hat{j} - 2y\hat{k}) \left[z \right]_0^{4-2x-2y} dy \, dx$$

$$= \int_{x=0}^{2} \int_{y=0}^{2-x} \left[(4-2x-2y)\hat{j} - (8y - 4xy - 4y^2)\hat{k} \right] dy \, dx$$

$$= \int_{x=0}^{2} \left[(4y - 2xy - y^2)\hat{j} - (4y^2 - 2xy^2 - \tfrac{4}{3}y^3) \right]_0^{2-x} dx$$

$$= \int_{x=0}^{2} \left[4(2-x) - 2x(2-x) - (2-x)^2 \right] \hat{j} - \left(4(2-x)^2 - 2x(2-x)^2 - \tfrac{4}{3}(2-x)^3 \right) \hat{k} \; dx$$

$4-4x+x^2$

$$= \left[4\frac{(2-x)^2}{-2} + 2\left(2x^2 - \frac{x^3}{3}\right) - \frac{(2-x)^3}{-3} \right]_0^2 \hat{j} - \left[4\frac{(2-x)^3}{-3} - 2\left(2x^2 - \tfrac{4}{3}x^3 + \frac{x^4}{4}\right) - \tfrac{4}{3}\frac{(2-x)^4}{-4} \right]_0^2 \hat{k}$$

$$= \left[-2(2-x)^2 - 2\left(x^2 - x^3/3\right) + \tfrac{1}{3}(2-x)^3 \right]_0^2 \hat{j} - \left[\tfrac{-4}{3}(2-x)^3 - 2\left(2x^2 - \tfrac{4}{3}x^3 + \tfrac{x^4}{4}\right) + \tfrac{1}{3}(2-x)^4 \right]_0^2 \hat{k}$$

$$= \left[\left(0 - 2(4 - 8/3) + 0\right) - \left(-8 - 0 + \tfrac{8}{3}\right) \right] \hat{j} - \left[\left(0 - 2\left(8 - \tfrac{32}{3} + 4\right) + 0\right) - \left(-\tfrac{32}{3} - 0 + \tfrac{16}{3}\right) \right] \hat{k}$$

$$= \left(-8 + \tfrac{16}{3} + 8 - 8/3\right)\hat{j} - \left(-16 + \tfrac{64}{3} - 8 + \tfrac{32}{3} - \tfrac{16}{3}\right)\hat{k} = \left(\tfrac{8}{3}\hat{j} - \tfrac{8}{3}\hat{k}\right)$$

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Gauss Divergence Thm :

Statement :

If $\vec{F}$ is a vector point function which is continuous and differential at point in region V bounded by closed surface S in a space.

then $$\iiint_V \nabla \cdot \vec{F} \, dv = \iint_S \vec{F} \cdot \hat{n} \, ds$$

where $\hat{n}$ is the outward (positive) unit normal vector to the surface S.

Ex : Using Gauss Divergence Thm. evaluate $\iint_S \vec{F} \cdot \hat{n} \cdot ds$

where $\vec{F} = 4xz\,\hat{i} - y^2\,\hat{j} + yz\,\hat{k}$ and S is the surface of the cube bounded by $x=0$, $x=1$, $y=0$, $y=1$, $z=0$, $z=1$

Soln :

Let $\vec{F} = 4xz\,\hat{i} - y^2\,\hat{j} + yz\,\hat{k}$

$$\therefore \nabla \vec{F} = \frac{\partial}{\partial x}(4xz) + \frac{\partial}{\partial y}(-y^2) + \frac{\partial}{\partial z}(yz)$$

$$= 4z - 2y + y$$

$$= 4z - y \quad \text{---(1)}$$

Now: using Gauss Divergence thm, we have

$$\iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V (\nabla \cdot \vec{F}) \, dv$$

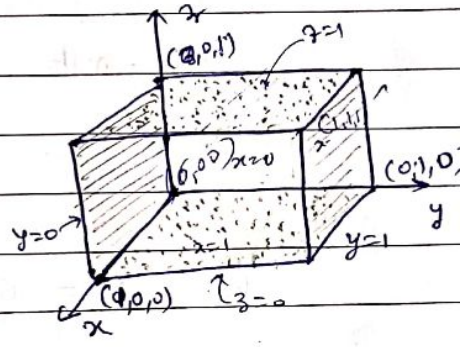
$$= \iiint_V (4z - y) \, dx \, dy \, dz \quad \text{---(2)}$$

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Here V is volume cube bounded by $x=0,\ x=1$
$y=0,\ y=1$
$z=0,\ z=1$

(2) $$\iint_S \bar{F}\cdot\hat{n}\,ds = \int_{x=0}^{1}\int_{y=0}^{1}\int_{z=0}^{1} (4z-y)\,dz\,dy\,dx$$



$$= \int_{x=0}^{1}\int_{y=0}^{1} \left[\frac{4z^2}{2} - y\cdot z\right]_0^1 \cdot dy\,dx$$

$$= \int_{x=0}^{1}\int_{y=0}^{1} (2-y)\,dy\,dx$$

$$= \int_{x=0}^{1} \left[2y - y^2/2\right]_0^1 dx$$

$$= \int_{x=0}^{1} 3/2\,dx \qquad = 3/2\,[x]_0^1 = 3/2$$

$$\therefore \iint_S \bar{F}\cdot\hat{n}\,ds = 3/2$$

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Ex: Using Gauss-Divergence thm evaluate $\iint_S \bar{A} \cdot \hat{n}\, ds$. where

S is the surface of the region bounded by $x^2+z^2=9$

$x=0$, $y=0$ $z=0$ and $y=8$.

Given that

$$\bar{A} = 6z\,\hat{i} + (2x+y)\,\hat{j} - x\,\hat{k}$$

Soln.

let $\bar{A} = 6z\,\hat{i} + (2x+y)\,\hat{j} - x\,\hat{k}$ ——— (1)

∴ By Gauss-Divergence thm, we have

$$\iint_S \bar{A}\cdot\hat{n}\, ds = \iiint_V \nabla\cdot\bar{A}\cdot dv \quad \text{——— (2)}$$

Here. $\nabla\cdot\bar{A} = \left(\frac{\partial}{\partial x}i + \frac{\partial}{\partial y}j + \frac{\partial}{\partial z}k\right)\cdot\left(6z\,\hat{i} + (2x+y)\,j - x\,k\right)$

$= \frac{\partial}{\partial x}(6z) + \frac{\partial}{\partial y}(2x+y) + \frac{\partial}{\partial z}(-x)$

$= 0 + 1 + 0 = 1$

also: V be the volume enclosed by surface $x^2+y^2=9$

plane $x=0$, $y=0$, $z=0$ and $y=8$

To cover whole volume

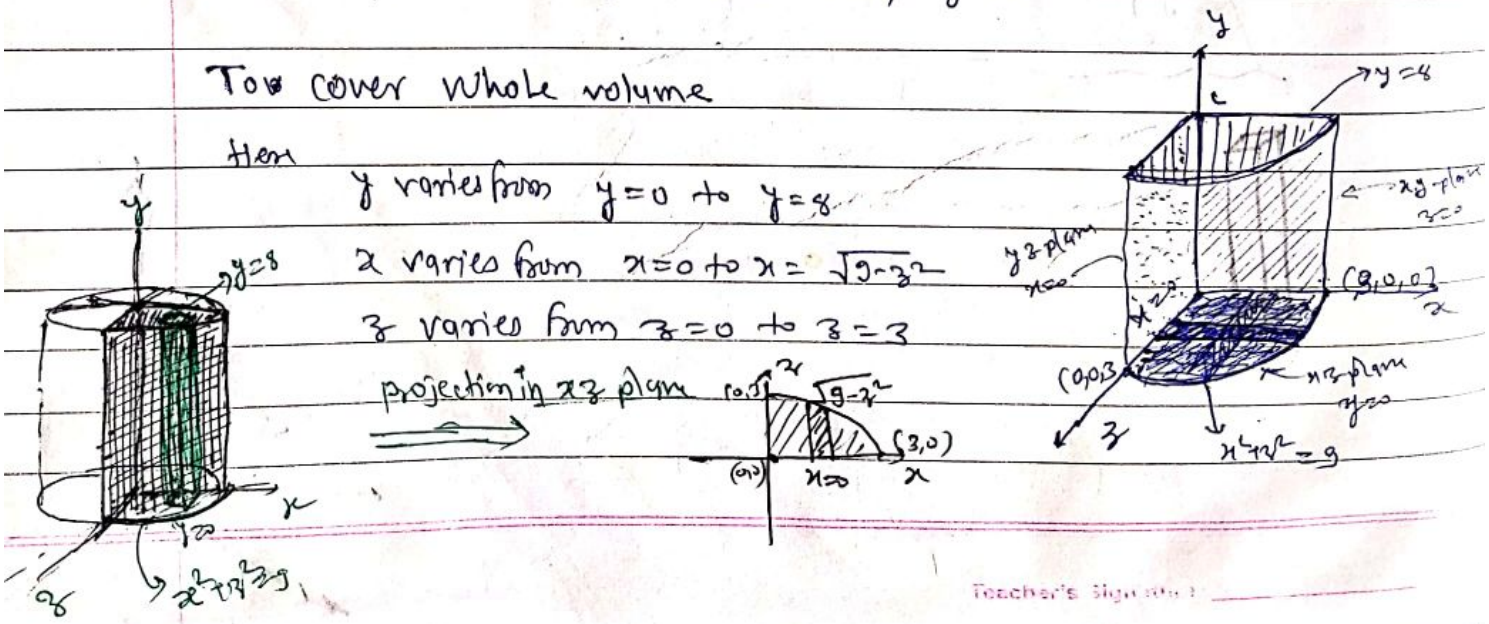
Here

y varies from $y=0$ to $y=8$

x varies from $x=0$ to $x=\sqrt{9-z^2}$

z varies from $z=0$ to $z=3$

projection in xz plane



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$\therefore$ (2) $\Rightarrow \iint_S \bar{A}\cdot\hat{n}\,ds = \int_{y=0}^{9}\int_{z=0}^{3}\int_{x=0}^{\sqrt{9-z^2}} dx\,dz\,dy$

$= \int_{y=0}^{9}\int_{z=0}^{3} \left[x\right]_0^{\sqrt{9-z^2}} dz\,dy$

$= \int_{y=0}^{9}\int_{z=0}^{3} \sqrt{9-z^2}\,dz\,dy = \int_{y=0}^{8}\left(\int_{z=0}^{3}\sqrt{3^2-z^2}\,dz\right)dy$

$= \int_{y=0}^{8}\left[\frac{z}{2}\sqrt{9-z^2} + \frac{(3)^2}{2}\sin^{-1}\left(\frac{z}{3}\right)\right]_0^3 dy$

$\because \int\sqrt{a^2-x^2} = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)$

$= \int_{y=0}^{8}\left[\left(0+\frac{9}{2}\sin^{-1}1\right) - \left(\frac{9}{2}\sin^{-1}0\right)\right]dy$

$= \frac{9}{2}\int_{y=0}^{8}\left(\frac{\pi}{2}-0\right)dy$

$= \frac{9}{4}\pi\int_{y=0}^{8} dy$

$= \frac{9}{4}\pi\cdot[8-0]$

$= 18\pi$

$\therefore \boxed{\iint_S \bar{A}\cdot\hat{n}\,ds = 18\pi}$

OR

$\iint_S \bar{A}\cdot\hat{n}\,ds = \iiint_V 1\,dv$

$= \iiint_V dx\,dy\,dz$

= (volume of cylinder in first octant)

$= \left[\frac{\pi r^2 h}{4}\right] = \frac{\pi(3)^2(8)}{4}$

$= \frac{72\pi}{4}$

$= 18\pi$

where h is height $= 8-4$

radius $r = 3$

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Ex: ⇒ Verify Divergence thm for

$$\vec{F} = (x^2-yz)\,\hat{i} + (y^2-zx)\,\hat{j} + (z^2-xy)\,\hat{k}$$

taken over the rectangular parallelopiped

$0 \le x \le a, \quad 0 \le y \le b, \quad 0 \le z \le c.$

Soln: let

$$\vec{F} = (x^2-yz)\,\hat{i} + (y^2-zx)\,\hat{j} + (z^2-xy)\,\hat{k} \qquad \text{---(1)}$$

$$\therefore \nabla\cdot\vec{F} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\cdot\left((x^2-yz)\,\hat{i} + (y^2-zx)\,\hat{j} + (z^2-xy)\,\hat{k}\right)$$

$$= \frac{\partial}{\partial x}(x^2-yz) + \frac{\partial}{\partial y}(y^2-zx) + \frac{\partial}{\partial z}(z^2-xy)$$

$$= 2x + 2y + 2z = 2(x+y+z).$$

Here: x varies from $x=0$ to $x=a$
y varies from $y=0$ to $y=b$
z varies from $z=0$ to $z=c$

$$\therefore \iiint_V \nabla\cdot\vec{F}\,dv = \int_{x=0}^{a}\int_{y=0}^{b}\int_{z=0}^{c} 2(x+y+z)\,dx\,dy\,dz$$

$$= 2\int_{x=0}^{a}\int_{y=0}^{b}\left[xz + yz + \frac{z^2}{2}\right]_0^c$$

$$= 2\int_{x=0}^{a}\int_{y=0}^{b}\left(cx + cy + \frac{c^2}{2}\right)dy\,dx$$

$$= 2\int_{x=0}^{a}\left[cxy + \frac{cy^2}{2} + \frac{c^2}{2}y\right]_0^b\,dx$$

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$$= 2 \int_{x=0}^{a} \left(bcx + c\frac{b^2}{2} + \frac{c^2}{2} b \right) dx$$

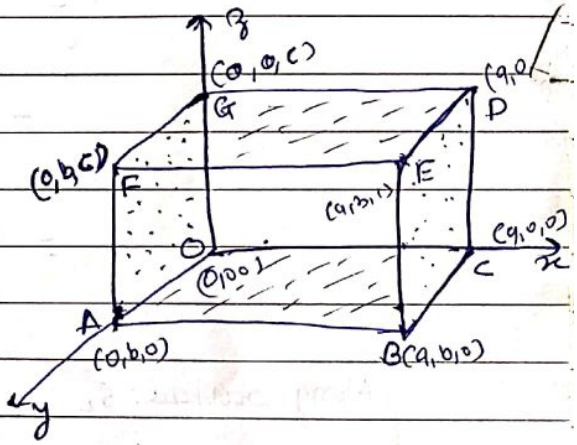
$$= 2 \left[bc\frac{x^2}{2} + \frac{cb^2}{2} \cdot x + \frac{c^2 b}{2} x \right]_0^a$$

$$= a^2bc + cb^2a + c^2ba$$

$$\iiint_V \nabla \cdot \vec{F} \, dV = abc\,(a + b + c) \qquad \text{---------- (2)}$$

Now: To evaluate surface integral

$$\iint_S \vec{F} \cdot \hat{n} \, ds.$$



The parallelopiped OABCDEFG having six surface. ~~Thus~~.

S_1 = surface OABC $(z=0)$

S_2 = surface DEFG $(z=c)$

S_3 = surface OAFG $(x=0)$

S_4 = surface BCDE $(x=a)$

S_5 = surface OCDG $(y=0)$

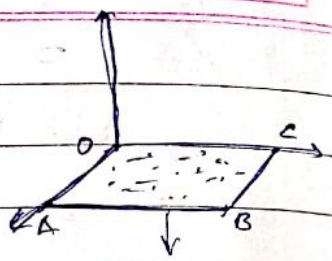
S_6 = surface ABEF $(y=b)$

$$\therefore \iint_S \vec{F} \cdot \hat{n} \, ds = \iint_{S_1} \vec{F} \cdot \hat{n} \, ds + \iint_{S_2} \vec{F} \cdot \hat{n} \, ds + \iint_{S_3} \vec{F} \cdot \hat{n} \, ds + \iint_{S_4} \vec{F} \cdot \hat{n} \, ds$$

$$+ \iint_{S_5} \vec{F} \cdot \hat{n} \, ds + \iint_{S_6} \vec{F} \cdot \hat{n} \, ds. \qquad \text{---------- (3)}$$

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Along surface S_1 : ie $z=0 \Rightarrow dz=0$



Here $\hat{n} = -\hat{k}$. (outward normal)

and $ds = \frac{dx\,dy}{|\hat{n}\cdot\hat{k}|} = \frac{dx\,dy}{|-\hat{k}\cdot\hat{k}|} = dx\,dy$

and x varies from $x=0$ to $x=a$
y varies from $y=0$ to $y=b$

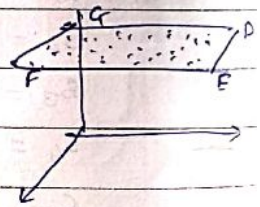
$$-\iint_{S_1} \vec{F}\cdot\hat{n}\,ds = \int_{x=0}^{a}\int_{y=0}^{b} (x^2\hat{i}+y^2-xy\hat{k})(-\hat{k})\,dx\,dy = \int_{x=0}^{a}\int_{y=0}^{b} xy\,dx\,dy$$

$$= \int_{x=0}^{a} \left[x\frac{y^2}{2}\right]_0^b dx$$

$$= \int_{x=0}^{a} \frac{xb^2}{2}\,dx = \left[\frac{b^2}{2}\cdot\left(\frac{x^2}{2}\right)\right]_0^a$$

$$= \frac{a^2b^2}{4}$$

Along surface S_2 : ie $z=c \Rightarrow dz=0$



Here $\hat{n} = \hat{k}$ and $ds = \frac{dx\,dy}{|\hat{n}\cdot\hat{k}|} = dx\,dy$

$$\iint_{S_2} \vec{F}\cdot\hat{n}\,ds = \int_{x=0}^{a}\int_{y=0}^{b} (\underset{-y(0)}{x^2\hat{i}}+\underset{-z(0)}{y^2\hat{j}}+(c^2-xy)\hat{k})\cdot\hat{k}\,dx\,dy$$

$$= \int_{x=0}^{a}\int_{y=0}^{b} (c^2-xy)\,dx\,dy$$

$$= \int_{x=0}^{a} \left[c^2y - \frac{xy^2}{2}\right]_0^b dx$$

$$= \int_{x=0}^{a} \left(c^2b - \frac{xb^2}{2}\right) dx = \left(c^2bx - \frac{b^2}{2}\cdot\frac{x^2}{2}\right)_0^a$$

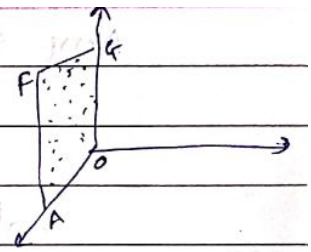
$$= abc^2 - \frac{a^2b^2}{4}$$

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Along S_3 : $x = 0$

and $\hat{n} = -\hat{i}$. $ds = \frac{dy\,dz}{|\hat{i}\cdot\hat{i}|} = dy\,dz$



and y varies from $y=0$ to $y=b$

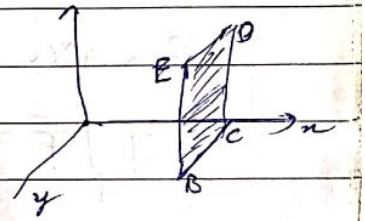
z varies from $z=0$ to $z=c$

$$\therefore \iint_{S_3} \vec{F}\cdot\hat{n}\,ds = \int_{z=0}^{c}\int_{y=0}^{b} (-yz\,i + y^2 j + z^2 k)\cdot(-\hat{i})\,dy\,dz$$

$$= \int_{z=0}^{c}\int_{y=0}^{b} yz\,dy\,dz = \int_{z=0}^{c} \frac{b^2 z}{2}\,dz = \frac{b^2 c^2}{4}$$

Along S_4 : $x = a$

and $\hat{n} = \hat{i}$ $ds = \frac{dy\,dz}{|\hat{i}\cdot\hat{i}|} = dy\,dz$



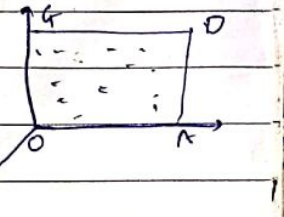
$\therefore 0 \le y \le b$, & $0 \le z \le c$

$$\therefore \iint_{S_4} \vec{F}\cdot\hat{n}\,ds = \int_{z=0}^{c}\int_{y=0}^{b} [(a^2 - yz)\,i + (y^2 - 3a)\,j + (z^2 - 4y)\,k]\cdot\hat{i}\cdot dy\,dz$$

$$= \int_{z=0}^{c}\int_{y=0}^{b} (a^2 - yz)\,dy\,dz = a^2 bc - \frac{b^2 c^2}{4}$$

Along S_5 : $y = 0$

$\hat{n} = -\hat{j}$ and $ds = \frac{dx\,dz}{(\hat{j}\cdot\hat{j})} = dx\,dz$



and $0 \le x \le a$, $0 \le z \le c$

$$\therefore \iint_{S_5} \vec{F}\cdot\hat{n}\,ds = \int_{x=0}^{a}\int_{z=0}^{c} [(x^2 - 0)\,i + (0 - 3x)\,j + (z^2 - 0)\,k]\,(-\hat{j})\cdot dx\,dz$$

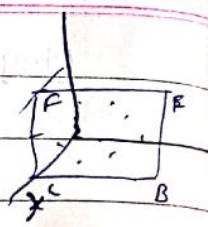
$$= \int_{x=0}^{a}\int_{z=0}^{c} (3x)\,dx\,dz = \frac{a^2 c^2}{4}$$

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Along S_6 : $y = b$

$\hat{n} = j$ $\quad ds = \frac{dx\,dz}{|\hat{n}\cdot j|} = \frac{dx\,dz}{(j\cdot j)} = dx\,dz$



and $0 \le x \le a, \quad 0 \le z \le c$

$$\therefore \iint_{S_6} \bar{F}\cdot\hat{n}\,ds = \int_{x=0}^{a}\int_{z=0}^{c} \left[(x^2 - bz)\,i + (b^2 - zx)\,j + (z^2 - xb)\,k\right]\cdot j \cdot dx\,dz$$

$$= \int_{x=0}^{a}\int_{z=0}^{c} (b^2 - zx)\cdot dz\,dx$$

$$= \int_{x=0}^{a} \left[b^2 c - \frac{c^2 x}{2}\right] dx = ab^2c - \frac{c^2 a^2}{4}$$

From eqn (2),

$$\iint \bar{F}\cdot\hat{n}\,ds = \frac{a^2 b}{4} + ab^2 c - \frac{a^2 b^2}{4} + \frac{b^2 c^2}{4} + a^2 bc - \frac{b^2 c^2}{4} + \frac{a^2 c^2}{4} - ab^2$$

$$= ab^2c + a^2bc + ab^2c$$

$$= abc\,(a^2 + b^2 + c^2) \quad \text{——— (3)}$$

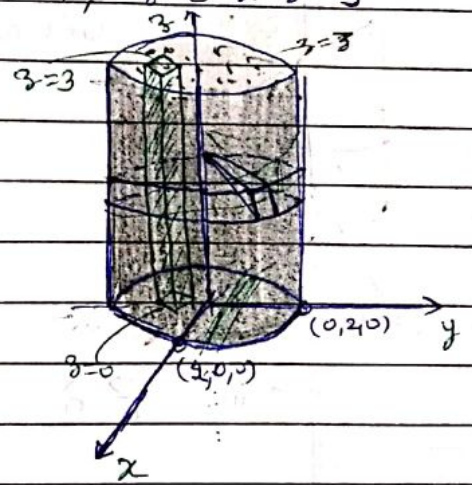
From eqn (2) & (3)

$$\iiint_V \nabla\cdot\bar{F}\,dV = \iint_S \bar{F}\cdot\hat{n}\,ds$$

Hence Divergence thm verified.

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Ex; Verify Gauss divergence thm for $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ taken over the region bounded by $x^2+y^2=4$, $z=0$ & $z=3$



Soln Let

$$\vec{F} = 4x\hat{i} - 4y^2\hat{j} + z^2\hat{k}$$

To evaluate $\iint_S \vec{F}\cdot\hat{n}\cdot ds$.

Since Total surface S of the cylinder consist of surface S_1 ($z=3$), S_2 ($z=0$) and surface S_3 (convex portion of cylinder $x^2+y^2=4$). Thus,

$$\iint_S \vec{F}\cdot\hat{n}\cdot ds = \iint_{S_1} \vec{F}\cdot\hat{n}\cdot ds_1 + \iint_{S_2} \vec{F}\cdot\hat{n}\cdot ds_2 + \iint_{S_3} \vec{F}\hat{n}\cdot ds_3 \quad \text{(1)}$$

For surface S_1 ($z=3$) [Circle with radius $r=2$ above xy plane]



unit outward normal to surface $S_1 \Rightarrow \hat{n} = \hat{k}$ (+ve)

$\therefore \vec{F}\cdot\hat{n} = (4x\hat{i} - 4y^2\hat{j} + z^2\hat{k})\cdot(\hat{k}) = z^2 = 3^2 = 9$

and $ds_1 = \frac{dx\,dy}{|n\cdot k|} = dx\,dy$

$\therefore \iint_{S_1} \vec{F}\cdot\hat{n}\,ds = \iint_{S_1} 9\cdot ds_1 = 9\iint_{S_1} ds_1 = 9\iint dx\,dy$

$= 9$ (Area of circle $x^2+y^2=4$)

$= 9(\pi 2^2) = 36\pi$.

For surface S_2 : ($z=0$) [Circle with radius $r=2$ in x-y-plane]



$\therefore$ unit outward normal to surface S_2 is

$\hat{n} = -\hat{k}$ (−ve)

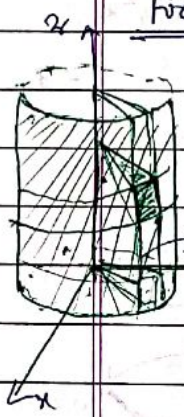
$\therefore \vec{F}\cdot\hat{n} = (4x\hat{i} - 4y^2\hat{j} + z^2\hat{k})(-\hat{k}) = -z^2 = 0$.

$ds_2 = \frac{dx\,dy}{|n\cdot k|} = \frac{dx\,dy}{|-1|} = dx$

$\therefore \iint_{S_2} \vec{F}\cdot\hat{n}\cdot ds_2 = 0$

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For surface : S_3 : [convex portion of cylinder $x^2+y^2=4$]

$\Rightarrow \phi = x^2+y^2-4$

$\therefore$ unit normal to the surface S_3 (ie $\phi = x^2+y^2-4$) is

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2x\hat{i}+2y\hat{j}}{\sqrt{4x^2+4y^2}} = \frac{2(x\hat{i}+y\hat{j})}{2\sqrt{x^2+y^2}}$$

$$= \frac{x\hat{i}+y\hat{j}}{\sqrt{4}} = \frac{1}{2}(x\hat{i}+y\hat{j})$$

Thus, $\vec{F}\cdot\hat{n} = (4x\hat{i}-2y^2\hat{j}+z^2\hat{k})\cdot\frac{1}{2}(x\hat{i}+y\hat{j})$

$= (2x^2-y^3)$

$$\iint_{S_3} \vec{F}\cdot\hat{n}\cdot ds = \iint_{S_3} (2x^2-y^3)\cdot dS_3$$

Changing into ~~polar~~ parametric form.

put $x = 2\cos\theta$ ~~$= 2\cos\theta$~~ (~~Hint $r=2$~~)

$y = 2\sin\theta$ ~~$= 2\sin\theta$~~

$dS_3 = 2\,d\theta\,dz = 2\,d\theta\,dz$

along surface S_3 : z varies from $z=0$ to $z=3$

θ varies from $\theta=0$ to $\theta=2\pi$

$$\iint_{S_3} \vec{F}\cdot\hat{n}\,ds = \int_{\theta=0}^{2\pi}\int_{z=0}^{3} \left[2\left(2^2\cos^2\theta\right) - \left(2\sin\theta\right)^3\right]\cdot 2\,d\theta\,dz$$

$$= \int_{\theta=0}^{2\pi}\int_{z=0}^{3} (8\cos^2\theta - 8\sin^3\theta)\,2\,d\theta\,dz$$

$$= 16\int_0^{2\pi} (\cos^2\theta - \sin^3\theta)\cdot[z]_0^3\,d\theta$$

$$= 48\int_0^{2\pi} (\cos^2\theta - \sin^3\theta)\,d\theta$$

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$$= 48\left[\int_0^{2\pi}\cos^2\theta\,d\theta - \int_0^{2\pi}\sin^3\theta\,d\theta\right]$$

odd function

$$= 48\left[\int_0^{2\pi}\cos^2\theta\,d\theta - 0\right]$$

$$= 48\int_0^{2\pi}\left(\frac{1+\cos 2\theta}{2}\right)d\theta$$

$$= \frac{48}{2}\left[\theta + \frac{\sin 2\theta}{2}\right]_0^{2\pi}$$

$$= \frac{48}{2}\cdot[2\pi - 0] = 48\pi.$$

Thus from eq^n (1)

$$\iint_S \vec{F}\cdot\hat{n}\,ds = 36\pi + 0 + 48\pi = 84\pi$$

(2)

Now, To evaluate : $\iiint_V \nabla\cdot\vec{F}\,dV$

as shown in fig. $z \to$ varies from $z=0$ to $z=3$

$y \to$ varies from $y = -\sqrt{4-x^2}$ to $y = \sqrt{4-x^2}$

$x \to$ varies from $x=-2$ to $x=2$

and

$$\nabla\cdot\vec{F} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\cdot\left(4x\hat{i} - 4y^2\hat{j} + z^2\hat{k}\right)$$

$$\boxed{\nabla\cdot F = 4 - 8y + 2z}$$

and $dv = dx\,dy\,dz$

$$\therefore \iiint_V \nabla\cdot\vec{F}\,dV = \int_{z=0}^{3}\int_{x=-2}^{2}\int_{y=-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4-8y+2z)\,dy\,dx\,dz$$

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$$= \int_{z=0}^{3} \int_{x=-2}^{2} \left[4y - 4y^2 + 2xy \right]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx\, dz$$

$$= \int_{z=0}^{3} \int_{x=-2}^{2} \left[\left(4\sqrt{4-x^2} - 4(4-x^2) + 2x\sqrt{4-x^2} \right) - \left(-4\sqrt{4-x^2} - 4\left(\sqrt{4-x^2}\right)^2 - 2x\sqrt{4-x^2} \right) \right] dx$$

$$= \int_{z=0}^{3} \int_{x=-2}^{2} \left(8\sqrt{4-x^2} + 4x\sqrt{4-x^2} \right) dx\, dz$$

$$= \int_{z=0}^{3} \int_{x=-2}^{2} (8+4x)\sqrt{4-x^2}\, dx\, dz$$

$$= \int_{z=0}^{3} (8+4z) \cdot \left[x/2\sqrt{4-x^2} + \frac{4}{2} \cdot \sin^{-1}(x/2) \right]_{-2}^{2} dz$$

$$= \int_{z=0}^{3} (8+4z) \left[\left(0 + 2\sin^{-1}(1) \right) - \left(0 + 2\sin^{-1}(-1) \right) \right] dz$$

$$= \int_{z=0}^{3} (8+4z) \left(2\pi/2 - 2(-\pi/2) \right) dz$$

$$= \int_{z=0}^{3} (8+4z) \cdot (2\pi)\, dz$$

$$= 2\pi \left(8z + 2z^2 \right]_0^3 = 2\pi \left[24 + 18 \right]$$

$$= 2\pi (42)$$

$$= 84\pi$$

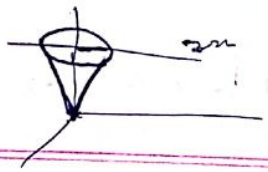
$$\boxed{\iiint_V \nabla \cdot \vec{F} \cdot dv = 84\pi} \quad \text{---------- (3)}$$

From (2) & (3)

$$\iint \vec{F} \cdot \hat{n} \cdot ds = \iiint \nabla \vec{F} \cdot dv$$

Hence Gauss divergence theorem verified

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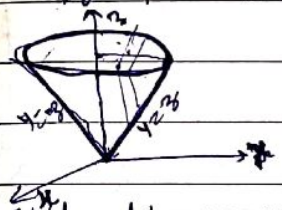


Hw

using Gauss-divergence thm

(1) Evaluate $\iint_S \vec{F}\cdot\hat{n}\,ds$ over the surface of the region above xy-plane bounded by the cone $z^2 = x^2+y^2$ and the plane $z=4$

If $\vec{F} = 4xz\,i + xyz^2\,j + 3z\,k$



z varies from $z=0$ to $z=4$

$\Rightarrow$ x varies from: $x = -\sqrt{z^2-y^2}$ to $x = \sqrt{z^2-y^2}$

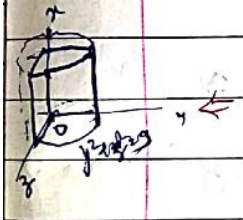
y varies from: $y=z$ to $y=-z$

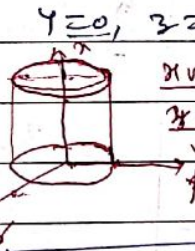
Ans = 320π

(2) State divergence thm and Hence evaluate $\iint_S \vec{F}\cdot\hat{n}\,ds$

Where $\vec{F} = 2x^2y\,\hat{i} - y^2\hat{j} + 4xz^2\hat{k}$

when S is closed surface of the region in the first octant bounded by cylinder $y^2+z^2=9$ and the planes $x=0, x=2$





$y=0, z\geq 0$

x varies fm: $x=0$ to $x=2$

z varies fm: $z=0$ to $z=\sqrt{9-y^2}$

y varies fm: $y=0$ to $y=3$

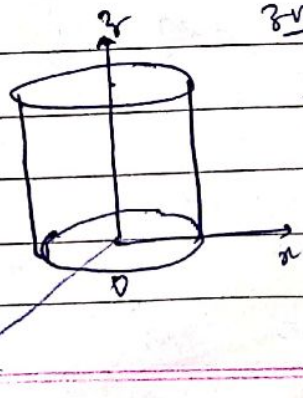
Ans 180.

(3) Use Divergence thm to evaluate $\iint_S \bar{A}\cdot\hat{n}\,ds$

Where $\bar{A} = x^2\,i + y^3\,j + z^3\,k$ and S is the surface of sphere $x^2+y^2+z^2 = a$

Ans $\frac{12\pi a^5}{5}$

(4) Use ~~verify~~ the divergence thm for $\vec{F} = 4x\,\hat{i} - 2y^2\,j + z^2k$ taken over the region bounded by the cylinder $x^2+y^2=4$, $z=0$ & $z=3$



z varies fm: $z=0$ to $z=3$

y varies fm: $y=-\sqrt{4-x^2}$ to $y=\sqrt{4-x^2}$

x varies fm: $x=-2$ to $x=+2$

Ans 84π

Teacher's Signature

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